

Morita Coefficients, Incidence, and Semistable Refinement

A supported comparison for a subdivided annulus

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Abstract

We formulate a supported Morita comparison for an explicit modification of a thickness-two nodal model. The modification introduces an exceptional projective line and two thickness-one nodes. We compute its charts and identify the complete inverse-image support, including the smooth part of the exceptional component. Using standard generation and recollement mechanisms for actual rational unipotent coefficients on the algebraic normal pair, we establish unit generation, a finite-descent generator, explicit link reconstruction, closure under the supported operations, and duality of the attaching datum. We also realize its unit and localization projectors by geometric diagonal kernels. A K_0 obstruction shows that the full exceptional constructible category cannot itself be refinement invariant. We also identify the modification as an fs log blow-up and apply Fujiwara–Kato acyclicity to obtain an actual finite-coefficient logarithmic Morita comparison, before forgetting base inertia. For geometric rational unipotent coefficients we identify the actual nearby image on the exceptional curve by an intrinsic costalk-vanishing condition, and identify its actual complete-support partner by nearby Verdier duality. This gives a module-valued star comparison and its supported partner for the single modification. For each fixed thickness $xy = t^m$, we give pairwise supported Morita comparisons for integer cut models over the same base. Their branchwise coefficient categories have geometric generators, and the specified proper maps compare ambient attaching maps, localization triangles, Verdier partners and minimal diagonal kernels. The coefficient passage is constructed in selected adic systems before rationalization. On the common rational unipotent source, an explicit one-modification bridge identifies these maps with the companion paper’s local Kummer realization and trace pairing, with node support interpreted as a costalk. For a fixed coarse support index we organize the fibrewise-cartesian ambient categories, localization, tensor, duality and minimal kernel actions into a coherent refinement diagram, and extend the local realization bridge naturally to that diagram. An unrestricted support/correspondence profile remains a separate target. Neither this target nor arbitrary model or base-change independence is claimed as proved. The logarithmic finite-coefficient comparison and the geometric rational nearby comparison have different sources; base inertia, analytic realization and integral supported refinement are not identified between them.

1 The supported comparison and its scope

Let R be a complete strictly henselian discrete valuation ring with uniformizer t , algebraically closed residue field k of characteristic p , and fraction field K . Fix $\ell \neq p$ and a complete algebraically closed valued extension Ω/K . Put

$$X = \operatorname{Spec} R[x, y]/(xy - t^2), \quad Y = V(x, y, t), \quad \mu : X' = \operatorname{Bl}_{(x, t)} X \longrightarrow X.$$

We keep the base R fixed. Thus the first example works also when $p = 2$; no claim about a tame quadratic base extension is required. The singular model X is a toroidal nodal model of thickness two, not the regular thickness-one model of the preceding local-realization paper.

Target 1.1 (The general supported refinement problem). Construct from intrinsically characterized actual nearby-admissible geometric coefficient categories a cellular Morita datum $\mathcal{C}(X, Y)$ and $\mathcal{C}(X', \mu^{-1}Y)$, together with a comparison induced by μ . Prove that the comparison is a Morita equivalence after the specified cellular descent, compatible with restriction, costalk, the localization attaching maps and Verdier duality. Identify its stratumwise cochain algebras and dualizing bimodules with the local realizations wherever the latter's hypotheses have been verified. Prove compatibility with compositions of the allowed modifications and finite geometric descent, and with a diagonal unit and the basic normal-pair correspondence.

The intended invariant is the entire coefficient datum on the stratified model: the stratum categories and their A^* -algebras, the supported $A^!$ -partners, actual restriction/costalk/localization functors and attaching cells, and normal inertia. Supported functors may have different source and target support categories; they are not all endomorphisms of a single coefficient category. In this sense the intended comparison has the schematic form

$$\mu_*^{\text{cell}} \mathcal{C}_{\Sigma_{X'}} \simeq_{\text{Mor}} \mathcal{C}_{\Sigma_X}.$$

Here Σ records actual strata and their incidence geometry. Theorem 10.11 proves pairwise equivalences for integer-cut models of a fixed $xy = t^m$, with the stated geometric rational coefficients and complete inverse-image supports. It includes the specified module, support and attachment maps. It does not establish the displayed schematic formula as an equivalence of fully coherent profiles; that stronger construction is Target 10.12.

The local finite-coefficient inputs are the explicit geometry and Nakayama's Kummer calculation. Lemma 4.1 constructs the selected adic comparisons; Theorem 10.6 proves geometric generation; Lemma 10.9 constructs the algebra and supported module mates; and Lemma 10.10 compares arbitrary retained attaching maps. These supply the pairwise theorem and Corollary 10.13. The logarithmic finite-coefficient comparison is a separate application of Fujiwara–Kato acyclicity. Proposition 13.1 connects the one-modification comparison to the companion paper's already proved rational unipotent interface, including the specified coefficient functor, module actions and support types. Theorem 11.3 organizes these maps on a fixed coarse support index; Corollary 13.3 places the coefficient interface in the same diagram. In this paper “canonical” refers to this specified cut/support system, not to a higher model-independent pseudofunctor on arbitrary correspondences.

Inputs and contribution. Fujiwara–Kato log-blow-up acyclicity and Nakayama's constant nearby calculation are external inputs, in the form recorded by Illusie [3]. Achinger–Ogus describe exterior cohomology and connecting/monodromy data [6, PDF version, §6.3]; we use this as background, not as an adic or module comparison input. Scherotzke–Sibilla–Talpo's Morita invariance concerns coherent parabolic coefficients [7], not the constructible coefficients used here. Recollement, thick generation from finite filtrations, one-generator Morita reconstruction and transport of an operation diagram along equivalences are standard mechanisms.

The geometric comparisons supplied here are the actual two-branch Kummer maps with their coefficient actions, the intrinsic admissible nearby image on the complete exceptional support, and the comparison of arbitrary retained attachments across coarse fibres. We organize these specified maps, their supported partners and the minimal kernel diagrams over a fixed coarse support index, and identify the common coefficient interface with the fixed companion manuscript. This is the contribution claimed, without a priority claim for the standard mechanisms or for general model independence.

Kummer classes and Tate twists. For an invertible coordinate z , the notation $[z]$ means its actual Kummer class in $H^1(-, \Lambda_r(1))$ at finite precision and in $H^1(-, Q(1))$ after the

specified adic passage and rationalization. Calling this a degree-one generator means the induced isomorphism $Q(-1) \rightarrow H^1(-, Q)$ in the rank-one situations below, not a scalar basis of untwisted cohomology. The Kummer morphism is $Q \rightarrow Q(1)[1]$. Identities such as $[w] = -[z]$ are identities in twisted cohomology and of these induced maps. We do not choose a trivialization $Q(1) \simeq Q$; compatible roots of the uniformizer used for coordinate transport do not constitute such a trivialization.

2 A modification that changes the model

Proposition 2.1 (Charts and the exceptional support). *The modification μ is proper and is an isomorphism on the generic fibre. It has affine charts*

$$U_x = \operatorname{Spec} R[x, v]/(xv - t), \quad y = xv^2, \quad U_t = \operatorname{Spec} R[u, y]/(uy - t), \quad x = tu.$$

On their overlap $u = v^{-1}$. The complete scheme-theoretic inverse image of Y is a reduced exceptional curve $E \simeq \mathbf{P}_k^1$. The special fibre of X' has two nodes q_0, q_∞ on E , and $E^\circ = E \setminus \{q_0, q_\infty\} \simeq \mathbf{G}_{m,k}$.

Proof. The ring $B = R[x, y]/(xy - t^2)$ embeds in $K[x, x^{-1}]$ by $y = t^2/x$. The x -chart of the blowup is the subalgebra $B[t/x]$ of its fraction field. Setting $v = t/x$ gives $xv = t$ and $y = xv^2$. This identifies it with the displayed R -flat domain; using the subalgebra description removes the spurious torsion that would result from retaining only the uncanceled equation $x(y - xv^2) = 0$. Similarly the t -chart is $B[x/t]$, with $u = x/t$ and $uy = t$. The two ratio coordinates are inverse on the overlap. Properness follows from the projective Rees construction, and (x, t) is the unit ideal after inverting t .

On U_x the ideal (x, y, t) pulls back to (x) ; on U_t it pulls back to (y) . These give $\operatorname{Spec} k[v]$ and $\operatorname{Spec} k[u]$, glued by $uv = 1$. Thus the inverse image is the reduced $\mathbf{P}^1$, with no support components omitted. The special fibres have equations $xv = 0$ and $uy = 0$. Their two intersections are $q_0 = (x, v)$ and $q_\infty = (u, y)$. These equations also verify that the new nodes have thickness one and that the models are regular there. $\square$

Let $T =]Y[_X$ denote the tube in the generic fibre of the t -adic completion. Normalize the valuation by $w(t) = 1$.

Proposition 2.2 (The full tube under subdivision). *The generic-fibre identification carries $]\mu^{-1}Y[_{X'}$ onto T . In the x coordinate,*

$$T = \{ |t|^2 < |x| < 1 \}.$$

The tubes of q_0, E°, q_∞ partition T by the conditions

$$0 < w(x) < 1, \quad w(x) = 1, \quad 1 < w(x) < 2,$$

respectively. The middle domain has coordinate $u = x/t$ with $|u| = 1$ and reduction in $\mathbf{G}_{m,k}$.

Proof. Specialization commutes with the formal morphism, so membership in the tube of the complete inverse image is equivalent to specialization in Y . Alternatively, $|x| < 1$ and $|y| < 1$ give $|t|^2 < |x| < 1$ directly. On U_x , specializing to q_0 requires $|x|, |v| < 1$ and $xv = t$, giving the first interval. On U_t , specializing to q_∞ requires $|u|, |y| < 1$ and $uy = t$, giving the last interval. Specialization to E° requires u and u^{-1} integral, hence $|x| = |t|$. $\square$

Remark 2.3. This partition is not an open analytic cover. In particular it supplies no ordinary open-cover Čech descent proof. The middle domain cannot be discarded, and compact-support duality for boundaryless open annuli cannot be copied onto it without checking its support convention. The rank-zero relative normal direction on E° does not erase its residue $\mathbf{G}_m$ geometry or the monodromy of its coefficients.

3 A logarithmic refinement comparison already available

Proposition 3.1 (The modification is an fs log blow-up). *Equip X with the chart $P \rightarrow R[x, y]/(xy - t^2)$, where*

$$P = \langle a, b, c \mid a + b = 2c \rangle, \quad a \mapsto x, \quad b \mapsto y, \quad c \mapsto t.$$

Then P is fine, saturated, sharp, with torsion-free groupification. The ordinary blow-up μ in Proposition 2.1, equipped with its displayed chart log structures, is the fs log blow-up of $J = (a, c) \subset P$. The induced morphism

$$p : E^{\log} \longrightarrow Y^{\log}$$

on the complete inverse-image support is again that log blow-up after strict base change. The logarithmic structures here are induced from the models, rather than the trivial structures on their underlying schemes.

Proof. Represent $a = (1, 0)$, $c = (0, 1)$ and $b = (-1, 2)$ in $\mathbf{Z}^2$. Then

$$P = \{(m, n) \in \mathbf{Z}^2 : n \geq 0, 2m + n \geq 0\}.$$

Indeed for $m \geq 0$ write $(m, n) = ma + nc$; for $m < 0$ write it as $(-m)b + (n + 2m)c$. This proves saturation as well as the asserted presentation and groupification.

The log blow-up chart where a generates the ideal has monoid $P + \mathbf{N}(c - a) = \mathbf{N}a + \mathbf{N}(c - a)$, with $c = a + (c - a)$ and $b = a + 2(c - a)$. It is already saturated and free. The other chart has monoid $P + \mathbf{N}(a - c) = \mathbf{N}(a - c) + \mathbf{N}b$, where $c = (a - c) + b$ and $a = 2(a - c) + b$. Both changes of lattice basis have unimodular determinant. On imposing $c \mapsto t$ their rings are exactly $xv = t$ and $uy = t$. Thus the saturation in the definition of fs log blow-up introduces no further modification. The chart construction and universal property are those of [3, §6.1].

Give Y the pulled-back log structure, so $Y^{\log} \rightarrow X^{\log}$ is strict. Its fs fibre product with X' has the ordinary underlying fibre product and the induced log structure. Proposition 2.1 identifies the underlying scheme with the full reduced E . The universal property of invertibility of the pulled-back ideal shows that this base change is the same log blow-up on $Y^{\log}$. $\square$

Theorem 3.2 (A finite-coefficient logarithmic Morita comparison). *Let $\Lambda_r = \mathbf{Z}/\ell^r$. For either $f = \mu$ or $f = p$, use the actual Kummer-étale topoi of its source and target. Then*

$$F \longrightarrow Rf_* f^* F$$

is an equivalence for every $F \in D^+(-_{\text{ket}}, \Lambda_r)$. Consequently f^ is fully faithful on this category. In particular, putting*

$$\mathbf{T}_r(Z) = \text{thick}_{D(Z_{\text{ket}}, \Lambda_r)}(\Lambda_r),$$

actual pullback induces a Morita equivalence $\mathbf{T}_r(Z) \simeq \mathbf{T}_r(Z')$ and identifies the derived endomorphism algebras of their units by the actual pullback map. For $r = 1$, $\mathbf{T}_1(Z)$ is also the geometrically described category of bounded complexes with finite unipotent locally constant cohomology on the indicated Kummer-étale topos.

Proof. Proposition 3.1 verifies the fs log blow-up hypothesis. The ideal is coherent and nonempty at every geometric stalk; where a generator becomes a unit the blow-up is simply the identity. The Fujiwara–Kato acyclicity theorem in [3, Theorem 6.2] therefore applies, and states precisely that the displayed adjunction unit is invertible for these bounded-below

torsion complexes. It is stronger than an equality of global cohomology groups. The unit belongs to the actual derived adjunction on complexes of sheaves; its being a quasi-isomorphism makes it an equivalence in the stable localization as well. Adjunction now gives, on the specified objects,

$$\mathrm{Map}(f^*F, f^*G) \simeq \mathrm{Map}(F, Rf_*f^*G) \simeq \mathrm{Map}(F, G).$$

Pullback is exact, sends the unit to the unit, and preserves retracts. Its fully faithful image on the thick unit subcategory is closed under finite cones and retracts and contains the target unit; hence it is exactly the target thick unit subcategory. The induced algebra map is an equivalence since it is this same comparison on the generator. For $r = 1$, finite locally constant sheaves are finite-dimensional representations of the topos fundamental group on each connected component. The filtration, kernel/cokernel and Postnikov argument of Proposition 4.4 applies verbatim to unipotent locally constant $\mathbf{F}_\ell$ -sheaves. This gives the geometric description independently of the thick-subcategory notation. For $r > 1$ we make no such assertion for every finite module: the theorem uses the explicitly selected finite-cell subcategory $\mathcal{T}_r$. $\square$

Corollary 3.3 (What coherence follows at this stage). *These comparisons commute with derived coefficient reduction on $\mathcal{T}_r$, preserve tensor products and duals of these finite-cell objects, and are compatible with compositions of actual log blow-ups. Precisely, put*

$$\mathcal{T}_{\mathrm{sys}}(Z) = \varprojlim_{r, -\otimes_{\Lambda_{r+1}}^L \Lambda_r} \mathcal{T}_r(Z).$$

This is a limit of stable categories. An object has specified reduction equivalences and all their composition cells; no uniform amplitude is implicit. Pullback induces an equivalence of these system categories. This is not an assertion about the ordinary inverse-limit sheaf.

Proof. Inverse image on module sheaves is exact and strong monoidal; its derived extension-of-scalars comparison is the usual associativity of tensor. Reduction takes the unit to the unit and therefore preserves the thick subcategories. Their objects are dualizable, being built from the unit by finite cones and retracts, so the same inverse image preserves their duals and evaluation maps. For g followed by f , the unit of the composite adjunction is the composite of their units, with the canonical composition cell for inverse image. This proves compatibility of the specified equivalences, not merely the existence of some equivalence for each modification. For the limit statement, the transition is derived scalar extension; it preserves the unit, finite cones and retracts, hence $\mathcal{T}_r$. For a chain of coefficient maps its comparison with pullback is the associativity map of the same derived tensor product. The unit and associativity identities for tensor define a natural transformation of the entire diagrams of categories. Its components are equivalences by Theorem 3.2; taking their categorical limit is therefore an equivalence. This proof does not use a sheafwise limit or assert coefficient base change for arbitrary direct images. $\square$

Corollary 3.4 (The ordinary direct-image shadow). *Write $\varepsilon_Z : Z_{\mathrm{ket}} \rightarrow \underline{Z}_{\mathrm{ét}}$ for the morphism forgetting the logarithmic site and $\underline{f}$ for the underlying scheme map. The actual log-refinement unit induces*

$$R\varepsilon_{Z,*}F \xrightarrow{\sim} R\underline{f}_*R\varepsilon_{Z',*}f^*F.$$

For $f = p$ the right side is the direct image from the complete exceptional curve, with its logarithmic coefficient realization retained.

Proof. The natural square of forgetful and direct-image morphisms gives $\varepsilon_Z \circ f = \underline{f} \circ \varepsilon_{Z'}$. Apply $R\varepsilon_{Z,*}$ to the actual unit of Theorem 3.2 and use composition of derived right adjoints. This identifies the displayed map and its inverse, naturally in F . No non-strict Kummer-etale proper base-change assertion is needed for this argument. $\square$

This shadow is not a full-faithfulness assertion for the ordinary realization, and it is not a comparison of ordinary exceptional restrictions. Nor has $R\mathcal{E}_*$ here been identified with the neutralized relative nearby functor.

This is a genuine comparison for a nontrivial model modification, using an existing geometric theorem. Its scope is crucial. On $Y^{\log}$ the characteristic group is P^{gp} of rank two, and the base class c is still retained. Although $P^{\text{gp}}/\mathbf{Z}c$ has rank one, passing to a neutralized relative log fibre and identifying its coefficient and support realizations is an additional step. Also $E^{\log}$ is not ordinary $E_{\text{ét}}$: forgetting logarithmic inertia can destroy full faithfulness. No ordinary cellular Morita invariance, exceptional functor, Verdier-support comparison or common-refinement independence is obtained merely by erasing these qualifications. In particular tensor duality here is not a proof of Verdier duality with geometric supports.

4 Geometric coefficients and their generators

4.1 The ambient category and geometric input

Put $Q = \mathbf{Q}_\ell$. For the finite-type k -schemes below we use the stable category of constructible adic complexes, with rationalized coefficients, and its bounded subcategory $D_c^b(-, Q)$. Mapping objects, fibres and adjunctions are taken in this enhancement, not in its homotopy category alone. The scheme case of adic six operations and its enhanced version are the external foundations [4, 5]. We use their localization, projection and Verdier duality maps. These are not assertions about an analytic specialization morphism or an infinite-root exceptional functor.

The additional geometric inputs are the Kummer computation for $\mathbf{G}_m \hookrightarrow \mathbf{A}^1$ and codimension-one purity, recorded in [3, §6.4, Theorems 7.1–7.2]. They are stated there with invertible finite coefficients. In the uses below the constant computations have finite free cohomology in degrees zero and one (purity in degree two), with the usual compatible reductions; taking the adic system and then rationalizing gives the stated Q computations. No general interchange of adic limits with arbitrary sheaf cohomology is being assumed.

External hypotheses and coefficient passage. The base trait is complete noetherian and hence excellent; all model schemes used here are separated of finite type over it, and all support schemes are of finite type over the algebraically closed residue field. The integer ℓ is invertible throughout. Fujiwara–Kato is used only on the Kummer-étale site, with a coherent nonempty log ideal on an fs locally noetherian scheme and $D^+(-, \Lambda_r)$ coefficients, as checked in Theorem 3.2. No ordinary full faithfulness follows from that theorem. Nakayama’s constant comparison is used for the vertical fs log smooth charts verified in Lemma 9.1 and Proposition 10.1, first with Λ_r . Geometric nearby cycles are taken after geometric generic base change; forgetting their Galois action is not taking inertia invariants.

Ordinary proper base change is used only for the proper scheme maps and their cartesian support squares, on bounded constructible complexes. Proper nearby comparison uses the same proper model maps. Their exceptional comparisons are Verdier mates, with dualizing complexes relative to the geometric points. Gabber’s theorem is used in the geometric form recorded in [10, Corollary 3.2, Remark 3.3]; the finite-level constant source is bounded constructible of finite Tor dimension, a subclass of the stated bounded-above constructible domain.

For passing these constant comparisons to Q , form the systems of actual Λ_r complexes and maps in the constructible adic formalism of [4, 5]. The Kummer and purity calculations have compatible derived reductions and uniformly bounded finite free stalk cohomology. Their supported constant partners are obtained by the finite-level duality maps; finite constructible

Tor amplitude is preserved in this use. The resulting adic systems have conservative coefficient reductions, so the levelwise equivalences are adic equivalences; rationalization gives the Q comparisons. This does not mean an unrestricted sheafwise inverse limit commutes with every six operation. For other objects of the rational unipotent source, the comparison maps are already defined in that formalism. Their functors are exact (contravariantly for duality), and the locus where a given map is an equivalence is closed under finite cones and retracts. The proved generation by Q , using finite filtrations and bounded Postnikov towers below, is precisely the extension argument. It neither presupposes integral generation of every rational coefficient nor proves an integral cellular theorem.

Lemma 4.1 (The coefficient systems used in the comparisons). *For the constant coefficients on the gap models and their finite stratified supports, the nearby, branch and proper comparison maps used below define maps of constructible adic systems. Their algebra and module structures are those of these systems before rationalization. Reduction is conservative on the systems, on their commutative algebras, and on their module objects. The constant global cochain algebras on the allowed supports also have their specified derived reductions.*

Proof. Use the enhanced adic category

$$\mathcal{D}_a(Z) = \varprojlim_r \mathcal{D}(Z, \Lambda_r), \quad \Lambda_r = \mathbf{Z}/\ell^r,$$

with derived extension of scalars as transition, and then its bounded constructible part. This is the construction of [2, §1.2, (1.1)–(1.7), Proposition 1.2.1]; comparison with adic complexes is [2, Theorem 1.3.6]. The tensor structure and inverse image are levelwise, as is shriek direct image on the stated finite-type, invertible-coefficient scheme domain. Right adjoints are the adic right adjoints, not arbitrary sheafwise limits. In particular proper direct image is levelwise because it equals shriek direct image. Our schemes are noetherian, finite dimensional and separated of finite type over the excellent trait or its geometric points, and ℓ is invertible; these are within that scheme domain.

For clarity put $E_Z = - \otimes_{\Lambda_{r+1}}^L \Lambda_r$. In the geometric generic/special diagram defining nearby cycles, write j for the generic map and i for special inverse image. The coefficient comparison is the mate

$$c_r : E_X Rj_{*,r+1} \longrightarrow Rj_{*,r} E_\eta$$

of the composite

$$j_r^* E_X Rj_{*,r+1} \simeq E_\eta j_{r+1}^* Rj_{*,r+1} \xrightarrow{E_\eta \epsilon} E_\eta.$$

Applying i^* and its scalar-extension identification gives

$$\rho_r : E_{X_s} R\Psi \Lambda_{r+1} \longrightarrow R\Psi \Lambda_r.$$

All mates are taken in the same coefficient-parameterized enhanced categories. Tensoring the counits gives the counit for each finite arity, so c_r and ρ_r are lax symmetric-monoidal transformations, with substitution and composition cells inherited from those counits. At the unit they are commutative-algebra maps; on coefficients they are module maps. If $a : N \rightarrow Rj_{U,*} j_U^* N$ is an open unit, its naturality reads

$$a_r \rho_r = (Rj_{U,*} j_U^* \rho_r) c_{U,r} E(a_{r+1}),$$

where the inverse-image/scalar identifications are understood. For proper comparison $\theta : N_I \rightarrow Ru_* N_J$, the corresponding identity is $\theta_r \rho_{I,r} = (Ru_* \rho_{J,r}) c_{u,r} E(\theta_{r+1})$. Both identities

are pastings of the same unit/counit squares. For a supported evaluation $N^\vee \otimes W \rightarrow \omega(-1)[-2]$, the coefficient square is

$$\begin{array}{ccc} E(N_{r+1}^\vee \otimes W_{r+1}) & \longrightarrow & E\omega_{r+1}(-1)[-2] \\ \downarrow & & \downarrow \\ N_r^\vee \otimes W_r & \longrightarrow & \omega_r(-1)[-2]. \end{array}$$

Here $N_r^\vee$ denotes the nearby coefficient dual input in the pairing, not an assertion that all nearby sheaves are tensor dualizable. The left arrow is the two coefficient comparisons and the right arrow is dualizing coefficient change. Commutativity follows from evaluation and trace naturality. Their equivalence on the selected constants is checked below; it is not required to construct this square.

At precision r form $R\Psi\Lambda_r$, its actual open-unit maps, Kummer morphisms, cup products, and proper comparison. A coefficient map gives the natural transformation obtained by scalar extension and the same adjunction units. Successive coefficient maps compose by derived tensor associativity, so these maps define a section, rather than a list of unrelated objects. Its cartesianness is a separate check: on a smooth stratum it is the constant reduction map; at a node and a branch link it is the reduction of the unit and primitive Kummer class in Proposition 10.1. The underlying complexes there have finite free cohomology in $[0, 1]$, so derived reduction has these same reduced generators and the specified comparison is an equivalence. The finitely many strata are conservative on constructible complexes. The earlier thickness- one and thickness-two uses follow from Lemma 9.1; neither local calculation requires this adic lemma.

The algebra maps are defined by the finite-level cup product and unit, not lifted from cohomology. Their coefficient squares are the cup- product naturality squares. Limits of commutative algebra objects and of algebra/module pairs are created on their underlying diagrams. Thus the resulting cartesian section is an algebra, with its modules, in $\mathcal{D}_a$. A map between such objects is an equivalence exactly when all its underlying coefficient evaluations are; no faithfulness assertion about cohomology as an algebra is used.

For the global cochain assertion, ordinary recollement expresses $R\Gamma(S, H_S)$ as a finite homotopy pullback of the open-stratum cochains, node cochains and branch links. This statement applies at finite precision before any Morita reconstruction. The open terms are the actual cochains of $\mathbf{G}_m$, whose Kummer calculation has compatible free reduction in degrees zero and one; the other terms were just checked. Derived scalar extension preserves finite homotopy limits in a stable category. Applying it to this specified pullback, including its attaching arrows, proves the global reduction assertion. This is not inferred by commuting stalks with $R\Gamma$ or with an inverse limit.

For supported constants, form the finite-level Gabber evaluation maps and then exceptional restriction. Their reduction compatibility can be checked by localization: smooth pieces have the purity factor $(1)[2]$, and closed pieces are the duals of the bounded free link fibres. Finite duals, these localization triangles and their coefficient maps commute with reduction. Thus the supported systems and the comparison of their pairings are cartesian as well. Proper support comparisons then use the levelwise proper direct image above. This only proves the claim for the constructed constants and the finite operations on them.

Rationalize these systems. Other geometric rational unipotent coefficients are covered by Proposition 4.4: the already defined exact natural transformations extend equivalences from the unit to finite cones and retracts. Multilinear comparisons are extended one variable at a time. No integral lattice for every rational object, or general integral cellular equivalence, is inferred. $\square$

Definition 4.2 (Geometric unipotent coefficients). For a connected smooth stratum Z , a finite-rank lisse $\mathcal{Q}$ -sheaf is *unipotent* if it has a finite filtration by lisse subsheaves with con-

stant successive quotients. Let $\mathcal{U}^b(Z)$ be the full subcategory of $D_c^b(Z, Q)$ whose cohomology sheaves have this property. For a geometric point put $\mathcal{U}^b(Z) = \text{Perf}(Q)$.

This is an absolute geometric starting domain. For a relative version, one must specify a base category and replace constant quotients by its actual pullbacks. The relative claim is not proved by the results below.

Lemma 4.3 (The one-object Morita argument). *Let $\mathcal{D}$ be an idempotent-complete stable linear category, with object G and algebra $A = R\text{Hom}(G, G)$. The functor $R\text{Hom}(G, -)$ identifies $\text{thick}(G)$ with $\text{Perf}(A)$, using right A -modules by precomposition.*

Proof. On G the comparison of mapping objects is the regular-module Yoneda identification. Exactness, finite sums and retracts extend it in each variable to $\text{thick}(G)$, proving full faithfulness. The image contains A and is closed under finite cones, shifts and retracts; these generate exactly $\text{Perf}(A)$. $\square$

Proposition 4.4 (Geometric unit generation). *For the connected smooth strata used here,*

$$\mathcal{U}^b(Z) = \text{thick}_{D_c^b(Z, Q)}(Q_Z) \simeq \text{Perf}(A_Z), \quad A_Z = R\Gamma(Z, Q).$$

This category is stable, idempotent-complete, and closed under tensor products and lisse duals. Pullback between the specified smooth strata preserves it.

Proof. Lisse sheaves correspond to finite-dimensional continuous representations. Kernels and cokernels of maps between unipotent representations are unipotent: intersect a unipotent filtration with an invariant subspace and take its images in the quotient. Extensions are unipotent by concatenating filtrations. The corresponding lisse sheaves have the same properties. The long exact cohomology sequence of a cone therefore shows that $\mathcal{U}^b(Z)$ is stable; an idempotent summand has cohomology an idempotent summand of the original representation and remains in the category. Thus $\text{thick}(Q_Z) \subseteq \mathcal{U}^b(Z)$.

Conversely a unipotent sheaf is built from constant finite-dimensional sheaves by its actual short exact sequences and their extension triangles. A bounded complex with these cohomology sheaves is then built by its finite Postnikov tower. This proves the reverse inclusion without defining the category by generation. Tensor filtrations and the reversed dual filtration prove the tensor and dual assertions on sheaves; finite truncations prove them on bounded complexes. Pullback preserves the lisse filtration. Apply Lemma 4.3 for the Morita equivalence. $\square$

Remark 4.5. The argument concerns kernels of maps *between lisse sheaves*. Lisse sheaves are not a Serre subcategory of all constructible sheaves: $j_!Q_U \subset Q_Z$ for a proper dense open U need not be lisse. In particular unit generation on a stratum does not prove closure under supported extension into the whole curve.

4.2 Keeping finite descent

Proposition 4.6 (A generator with finite geometric descent). *Let a finite group Δ act on a specified scheme with invariant stratification, and let $\mathcal{C}$ be one of the geometrically defined categories below, invariant under pullback. If $\mathcal{C} = \text{thick}(G)$, then its category $\mathcal{C}^{h\Delta}$ of actual equivariant objects is generated by $G_\Delta = \text{Ind}_\Delta(G)$ and hence is Morita equivalent to $\text{Perf}(R\text{Hom}_{\mathcal{C}^{h\Delta}}(G_\Delta, G_\Delta))$. The underlying object of G_Δ is $\bigoplus_{\delta \in \Delta} \delta^*G$.*

Proof. Use the derived category of sheaves with their geometric descent data; finite induction and coinduction agree, as finite sums and products agree. For an equivariant object

F , the counit $\text{Ind For}(F) \rightarrow F$ has an equivariant section: sum the inverse descent isomorphisms into its components and divide by $|\Delta|$. Their composite is the identity, including the coherent descent structure. Since Q has characteristic zero this is valid even if ℓ divides $|\Delta|$. The underlying object of F lies in $\text{thick}(G)$; induction is exact and preserves retracts, so $\text{Ind For}(F)$ lies in $\text{thick}(G_\Delta)$. Its retract F does too. The reverse inclusion follows from the geometric category's stability. Lemma 4.3 finishes the proof. Equivalently these constructions take place in the homotopy-coherent descent category, where the finite induction monad is the same geometric descent monad. $\square$

For the quotient stack this is its actual equivariant coefficient category. The proposition retains representations that would be lost by using only the equivariant trivial object. It does not prove integral averaging or descent between different refinements.

5 The link of an actual normal pair

Let C be either $\mathbf{A}_k^1$ with $Z = \{0\}$, or $E = \mathbf{P}_k^1$ with $Z = \{0, \infty\}$. In both cases put $U = C \setminus Z \simeq \mathbf{G}_m$, with immersions $j : U \rightarrow C$ and $i : Z \rightarrow C$. Define the *actual link functor*

$$L = i^* Rj_* : \mathcal{U}^b(U) \longrightarrow D(Z, Q), \quad L_a = i_a^* Rj_* \quad (a \in Z).$$

Its definition uses a geometric direct image; it is not an exterior insertion operator.

Proposition 5.1 (The local Kummer link). *For every endpoint a , $B_a = L_a(Q_U)$ is perfect over Q , with $H^0(B_a) = Q$ and $H^1(B_a) = Q(-1)$ and no other cohomology. The actual restriction map*

$$\kappa_a : A_U = R\Gamma(U, Q) \longrightarrow B_a$$

is an equivalence of cochain algebras. For every $V \in \mathcal{U}^b(U)$ its coefficient version $R\Gamma(U, V) \rightarrow L_a(V)$ is an equivalence. In particular L takes values in $\text{Perf}(Q)^{\mathbb{Z}}$.

Proof. The geometric stalk of Rj_* is computed on the punctured strict henselization at a . The Kummer computation cited above gives the asserted local cohomology and the global cohomology of $\mathbf{G}_m$. The actual restriction is the identity in degree zero. In degree one, the Kummer class of the coordinate z has residue 1 at zero and -1 at infinity, where the local coordinate is z^{-1} . Consequently the same restriction map is an isomorphism in every degree. It is a map of cochain algebras because it is induced by restriction on the sites, so the cohomology calculation proves the asserted equivalence; no formality of either algebra is required.

The coefficient restriction is a natural transformation of exact functors, constructed before comparing their values. Its equivalence locus is closed under finite cones and retracts and contains Q_U . Proposition 4.4 therefore proves the assertion for all V . This also proves perfectness and boundedness of the actual link on the whole chosen category. $\square$

Proposition 5.2 (A geometric supported residue). *For $(\mathbf{A}^1, \{0\})$, the localization boundary*

$$H^1(\mathbf{G}_m, Q) \longrightarrow H_{\{0\}}^2(\mathbf{A}^1, Q)$$

is nonzero and is an isomorphism. With the purity normalization by the effective divisor $\{0\}$ it sends the Kummer coordinate class to its cycle class. For an arbitrary unipotent lisse coefficient V on U , localization also gives the canonical equivalence

$$i^! j_! V \simeq L(V)[-1].$$

Proof. Homotopy invariance gives $H^{>0}(\mathbf{A}^1, Q) = 0$; purity gives $i^!Q_{\mathbf{A}^1} = Q(-1)[-2]$. The localization long exact sequence therefore gives the first isomorphism. Kummer boundary and the cycle class of a parameter give the stated normalization, as in the cited codimension-one computation.

For the second assertion use the actual triangle $j_!V \rightarrow Rj_*V \rightarrow i_*L(V) \rightarrow$. The object $i^!Rj_*V$ is zero: for every W its maps from W are maps $i_*W \rightarrow Rj_*V$, equivalently maps $j^*i_*W \rightarrow V$, which vanish. Applying $i^!$ to this triangle identifies its connecting map $L(V) \rightarrow i^!j_!V[1]$ as an equivalence. This determines both the supported object and its attaching map. $\square$

Example 5.3 (A coefficient with nontrivial monodromy). Fix compatible Kummer roots of z and a generator of the relative $\mathbf{Z}_\ell(1)$. Let V be the actual rank-two local system from the continuous Kummer representation

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{on } Qe_1 \oplus Qe_2.$$

The integral lattice is stable, and its reductions factor through finite Kummer covers, so this specifies a geometric adic local system. The punctured-trait link is continuous inertia cohomology, computed by

$$L_0(V) \simeq [V \xrightarrow{T-1} V] \quad \text{in degrees } 0, 1,$$

after the generator choice trivializes the Tate line. This is the pro-cyclic resolution: the augmentation ideal of $\mathbf{Z}_\ell[[\mathbf{Z}_\ell]]$ is generated by the chosen $\gamma - 1$, with the two-term free resolution of the trivial module. For this tame coefficient wild inertia contributes no higher ℓ -cohomology; the prime-to- ℓ part is exact and acts trivially. The Kummer punctured-trait description is that of [3, §4.7]. Thus $H^0(L_0(V)) = Qe_1$ and $H^1(L_0(V)) = (V/(T-1)V)(-1)$, a one-dimensional space generated by the class of e_2 with the Tate line restored. At infinity the action is T^{-1} , not a second independently chosen monodromy. Proposition 5.2 gives

$$H^1(i_0^!j_!V) = Qe_1, \quad H^2(i_0^!j_!V) = (V/(T-1)V)(-1).$$

In particular the actual supported boundary remains nonzero for a nonconstant coefficient. The complex has been obtained from the geometric link and its localization map, not assigned as its definition.

6 Constructible reconstruction and operation closure

Definition 6.1 (The stratified source). Let $\mathcal{C}(C, Z)$ be the full subcategory of $D_c^b(C, Q)$ consisting of F for which $j^*F \in \mathcal{U}^b(U)$ and $i^*F \in \text{Perf}(Q)^Z$. These are conditions on restrictions of actual constructible complexes. No closure under exceptional operations is included in the definition.

Theorem 6.2 (Actual gluing and its Morita presentation). *There is an equivalence of stable categories between $\mathcal{C}(C, Z)$ and triples*

$$(V, W, \alpha), \quad V \in \mathcal{U}^b(U), \quad W \in \text{Perf}(Q)^Z, \quad \alpha : W \longrightarrow L(V).$$

The arrow and the compatibility of morphisms are homotopy coherent. Under this equivalence,

$$i^!F = \text{fib}(W \xrightarrow{\alpha} L(V)).$$

The category is generated by the actual constructible object

$$G_C = j_! Q_U \oplus \bigoplus_{a \in Z} i_{a,*} Q,$$

and is Morita equivalent to $\text{Perf}(A_C)$ with $A_C = R\text{Hom}_C(G_C, G_C)$.

Proof. For F set $V = j^* F$, $W = i^* F$, and let α be the restriction of the adjunction unit $F \rightarrow Rj_* j^* F$. Conversely form in the ambient stable sheaf category

$$F(V, W, \alpha) = Rj_* V \underset{i_* L(V)}{\overset{h}{\times}} i_* W.$$

Its two restrictions are V and W . Proposition 5.1 and the localization triangle for $Rj_* V$ show that it is bounded and constructible in the stated source. The canonical comparison from an original F to this pullback is an equivalence after j^* and i^* ; these functors are jointly conservative by localization. This proves reconstruction, naturally on objects and morphisms. Explicitly the mapping spectrum between two triples is the fibre of

$$\text{Map}_U(V, V') \oplus \text{Map}_Z(W, W') \longrightarrow \text{Map}_Z(W, L(V')), \quad (f, g) \longmapsto L(f)\alpha - \alpha'g.$$

Thus the comparison retains the required homotopies, not just commuting squares in a triangulated category. Applying i^* to $i_* i^! F \rightarrow F \rightarrow Rj_* V$ gives the stated fibre formula.

The geometric definition is stable and closed under retracts by its restriction conditions and Proposition 4.4. The triangle $j_! V \rightarrow F \rightarrow i_* W$ and that proposition show that every F is in $\text{thick}(G_C)$. Each summand of G_C satisfies the defining geometric conditions, proving the reverse inclusion. Apply Lemma 4.3. $\square$

The full derived endomorphism algebra has the following blocks, including the nonzero off-diagonal ones:

$$\begin{aligned} R\text{Hom}(j_! Q, j_! Q) &= A_U, & R\text{Hom}(j_! Q, i_{a,*} Q) &= 0, \\ R\text{Hom}(i_{a,*} Q, j_! Q) &= B_a[-1], & R\text{Hom}(i_{a,*} Q, i_{b,*} Q) &= \begin{cases} Q & a = b, \\ 0 & a \neq b. \end{cases} \end{aligned}$$

The first two formulas use $j_! \dashv j^*$; the third uses Proposition 5.2; disjoint closed supports give the last. The A_U -action on B_a is induced by the actual link functor and κ_a . Thus these are derived blocks with geometric composition, not merely a matrix of graded cohomology groups.

In the right-module Morita convention the link is

$$M \longmapsto M \otimes_{A_U}^L B_a, \quad M = R\text{Hom}_U(Q_U, V),$$

where B_a has its left A_U -action induced by L_a on endomorphisms. The natural evaluation comparison is an equivalence on Q_U , hence on its thick closure. Together with the mapping formula above this identifies the coefficient-valued attaching functor itself.

Proposition 6.3 (The supported operations really preserve the source). *For these pairs, $j^*, i^*, i^!, j_!, Rj_*, i_*$ preserve the corresponding categories $\mathcal{C}(C, Z)$, $\mathcal{U}^b(U)$ and $\text{Perf}(Q)^Z$. The global source is closed under tensor products and Verdier duality.*

Proof. The restriction assertions follow from the geometric conditions. For $j_! V$, $i_* W$ and $Rj_* V$ the triples of Theorem 6.2 are respectively

$$(V, 0, 0), \quad (0, W, 0), \quad (V, L(V), \text{id}).$$

Their membership follows from the proved perfectness of $L(V)$. The formula for $i^!$ is a fibre of perfect complexes. Tensor products are checked on the two strata, where Proposition 4.4 and ordinary finite-dimensional tensor products apply.

On U smooth-curve duality is $D_U V = V^\vee(1)[2]$, which preserves $\mathcal{U}^b(U)$; on Z it is ordinary Q -duality. The actual identities $D_C j_! = Rj_* D_U$ and $D_C i_* = i_* D_Z$ put the duals of all summands of G_C in $\mathcal{C}(C, Z)$. Exactness, retracts and the proved generation show that D_C preserves the whole category. Constructible Verdier biduality proves that it is an anti-equivalence. Thus support closure and duality are consequences, not defining assumptions. $\square$

Every construction is equivariant for geometric automorphisms of the pair, including permutation of endpoints. Consequently these closure statements also hold with finite geometric descent, using Proposition 4.6. This statement concerns a fixed pair with its actual action, not arbitrary comparison of atlases.

6.1 Duality of the whole attaching datum

Proposition 6.4 (Duality on reconstructed triples). *For a triple (V, W, α) put $C_\alpha = \text{fib}(\alpha)$. Its Verdier dual has triple*

$$(D_U V, D_Z C_\alpha, \alpha^D), \quad \alpha^D : D_Z C_\alpha \longrightarrow D_Z L(V)[1] \simeq L(D_U V).$$

The first arrow is the boundary of the dual of $C_\alpha \rightarrow W \rightarrow L(V)$. The equivalence on the right is canonical.

Proof. The restrictions of $D_C F$ are $D_U j^* F$ and $D_Z i^! F$, giving the two stated objects. Moreover actual Verdier exchange and localization give

$$D_Z L(V) = D_Z i^* Rj_* V \simeq i^! j_! D_U V \simeq L(D_U V)[-1].$$

Dualize the actual localization triangle of F . Its restriction to Z identifies the resulting unit with the displayed boundary, which therefore is the attaching map of $D_C F$. Biduality and the unit/counit identities give the coherence of this operation on morphisms. $\square$

For $F = Q_{A^1}$ the triple has $V = Q_U$, $W = Q$ and $\alpha : Q \rightarrow B_0$ the actual degree-zero unit. Its fibre is $Q(-1)[-2]$, recovering purity; setting $\alpha = 0$ would give a different costalk. This is a direct check that attaching data cannot be dropped from the comparison.

On the open stratum the cochain algebra and its supported partner are

$$A_U = R\Gamma(U, Q), \quad P_U = R\Gamma_c(U, Q) \simeq D_Q A_U(-1)[-2].$$

For a unipotent lisse V , geometric smooth-curve duality gives

$$R\Gamma_c(U, V) \simeq D_Q R\Gamma(U, V^\vee)(-1)[-2].$$

The comparison uses the actual evaluation and compact-support trace. Cup-product associativity makes evaluation balanced for the two A_U -actions; dualizing the regular actions gives the corresponding base-linear dualizing bimodule. The adjunction identities in Proposition 6.4 show how this partner enters the localization maps. These are statements about the algebraic pair; identification with analytic and relative-root supported realizations across the modification remains a separate geometric comparison. In particular we assert neither $A_U \simeq P_U$ nor a Calabi–Yau identification of the full algebra A_C with its linear dual.

6.2 Two intrinsically described support categories

Proposition 6.5 (The star and shriek pieces of the recollement). *Inside the proved ambient category define by geometric vanishing*

$$\mathcal{C}_* = \{F : i^!F = 0\}, \quad \mathcal{C}_! = \{F : i^*F = 0\}.$$

Then $Rj_* : \mathcal{U}^b(U) \simeq \mathcal{C}_*$ and $j_! : \mathcal{U}^b(U) \simeq \mathcal{C}_!$ are equivalences, with inverse j^* . Verdier duality exchanges these categories. The canonical map $j_!V \rightarrow Rj_*V$ has cone $i_*L(V)$ and retains the nonzero link. For $C = E$ proper, their integrated values are respectively $R\Gamma(U, V)$ and $R\Gamma_c(U, V)$.

Proof. In the reconstruction theorem $i^!F = 0$ says exactly that $\alpha : W \rightarrow L(V)$ is an equivalence, which identifies the triple with $(V, L(V), \text{id})$. The condition $i^*F = 0$ says $W = 0$ and gives $(V, 0, 0)$. These arguments give both essential surjectivity and the functors on maps; alternatively full faithfulness is the open-immersion adjunction. The Verdier exchanges and the displayed localization triangle were proved above. Properness of E and composition of direct images give the integrated identities. $\square$

These are intrinsic categories on a fixed algebraic pair, and each has $K_0 = \mathbf{Z}$. They are useful candidates for the star and supported pieces of a nearby realization. Membership of the actual nearby image requires a separate geometric calculation of its endpoint attaching maps. This calculation is carried out in Theorem 9.2, and the actual supported partner in Theorem 9.4. The present proposition itself is not a definition of nearby cycles.

Nor is $\mathcal{C}_*$ a tensor subcategory for the ordinary ambient tensor product. For $F = Rj_*Q_U$, its endpoint stalk is B_a . The attaching map of $F \otimes F$ is, after the canonical identifications, $B_a \otimes_Q B_a \rightarrow B_a$, the multiplication induced by restriction. Its source has nonzero $H^2 = Q(-2)$ whereas its target has zero H^2 . Thus it is not an equivalence and $F \otimes F \notin \mathcal{C}_*$. If tensor operations on a star realization are represented using the stratum tensor and Rj_* , their relation to geometric tensor operations must be separately proved; the ambient tensor cannot be substituted.

7 A minimal geometric kernel action

Here take $C = E = \mathbf{P}_k^1$. Let $\Delta : C \rightarrow C \times C$ be the diagonal and p_1, p_2 the projections. In the actual constructible category on the product consider just the kernels

$$K_{\text{id}} = \Delta_*Q_C, \quad K_U = \Delta_*j_!Q_U, \quad K_Z = \Delta_*i_*Q_Z.$$

They are allowed to be constructible along the diagonal, rather than lisse on the product. Set $\Phi_K(F) = Rp_{2,!}(p_1^*F \otimes^L K)$.

Proposition 7.1 (Unit and normal-pair projectors). *These kernels act on $\mathcal{C}(C, Z)$ by*

$$\Phi_{K_{\text{id}}} = \text{id}, \quad \Phi_{K_U} = j_!j^*, \quad \Phi_{K_Z} = i_*i^*.$$

Their kernel localization triangle induces the actual localization triangle of these functors. Under the Morita equivalence of Theorem 6.2 these actions are tensor with the bimodules $R\text{Hom}(G_C, \Phi_K G_C)$; the unit is the regular bimodule A_C .

Proof. For a constructible H on C , the diagonal projection formula gives $p_1^*F \otimes \Delta_*H = \Delta_*(F \otimes H)$. Since $p_2\Delta = \text{id}$ and Δ is proper, $\Phi_{\Delta_*H}(F) = F \otimes H$ with its actual natural map. Take $H = Q_C, j_!Q_U, i_*Q_Z$ and use the open and closed projection formulas. Proposition 6.3

proves preservation of the specified source. Applying the exact diagonal and kernel action to $j_!Q_U \rightarrow Q_C \rightarrow i_*Q_Z$ proves the triangle assertion, including its boundary map.

For any of these exact functors T , its action on endomorphisms gives a left A_C -action on $R\mathrm{Hom}(G_C, TG_C)$, commuting with the right action by precomposition. Evaluation constructs the natural tensor comparison; it is an equivalence on the generator and hence on all perfect modules. For $T = \mathrm{id}$ this is the regular bimodule and its identity action. $\square$

This proves a minimal kernel interface on the fixed algebraic pair. It does not prove compatibility of these kernels with refinement, nor closure or associativity for all relative-product convolution kernels.

8 Why the full exceptional category cannot be invariant

Proposition 8.1 (An obstruction to unrestricted refinement descent). *For the exceptional curve with its three strata,*

$$K_0(\mathcal{C}(\mathbf{P}^1, \{0, \infty\})) = \mathbf{Z}^3.$$

The bounded unipotent local coefficient category of one connected node, with constant simple coefficient, has $K_0 = \mathbf{Z}$; in particular this holds for unipotent continuous $\mathbf{Z}_\ell(1)$ representations in their geometric derived category. These two categories are not Morita equivalent.

Proof. For F take the Euler ranks of its restrictions to U , 0 , and ∞ . They define a homomorphism to $\mathbf{Z}^3$. The classes of $j_!Q_U, i_{0,*}Q, i_{\infty,*}Q$ map to its standard basis. Localization expresses $[F]$ as the sum of the open and closed contributions. On the open, finite unipotent filtrations and the Postnikov tower express every class as its Euler rank times $[Q_U]$; the point contributions have the same elementary property. Thus the three classes generate with no relations. The identical filtration argument at the old node, together with Euler rank one of its constant coefficient, gives $\mathbf{Z}$. Morita equivalence of these small idempotent-complete stable categories would preserve K_0 . $\square$

Even proper integration forgets the problem: $R\Gamma(E, i_{0,*}Q)$ and $R\Gamma(E, i_{\infty,*}Q)$ are both Q , but the mapping spectrum between the two disjoint skyscrapers on E is zero. Integration cannot be fully faithful on the constructed category.

Consequently $\mathcal{C}(C, Z)$ is a proved ambient recollement category, not yet the desired refinement-invariant source. Target 1.1 must use a *geometrically characterized nearby-admissible subdatum* and prove its reconstruction. Its characterization must retain the actual specialization maps and the required extension data. Defining it to be the image of the old category would prove neither an intrinsic criterion nor independence of the old model.

There is a further constraint: requiring that a refinement-invariant full stable subcategory of $\mathcal{C}(E, Z)$ contain the constant object on all of E and be closed under the ambient projectors $j_!j^*$ and each $i_{a,*}i_a^*$ forces it to contain all three generators above. Thus the required supported operations must be typed with their own support categories; they cannot all be endofunctors of a single category with the old node's K_0 . This is an obstruction to that formulation, not to a correctly typed multi-support Morita datum. The formulation of support transport is therefore part of the remaining theorem.

The costalks used in the recollement proof are those of the endpoints inside E . They are not the exceptional restriction along $E \hookrightarrow X'_s$. The following section constructs that additional comparison from the actual nearby-duality theorem and verifies admissibility by the actual Kummer attaching maps.

9 Actual nearby coefficients on the exceptional curve

Fix an algebraic closure $\bar{K} \subset \Omega$, and use geometric nearby cycles, with the base Galois action forgotten in this section. Identify $X'_\eta = X_{\bar{\eta}} = \mathbf{G}_{m, \bar{K}}$ by x . The source is the actual category $\mathcal{V} = \mathcal{U}^b(\mathbf{G}_{m, \bar{K}})$ of Proposition 4.4. Write

$$\kappa : E \hookrightarrow X'_s, \quad N(F) = \kappa^* R\Psi_{X'} F, \quad \mathcal{S}(F) = j^* N(F), \quad \alpha(F) : i^* N(F) \longrightarrow L(\mathcal{S}(F)).$$

Here α is the restriction of the actual open-immersion unit $N(F) \rightarrow Rj_* j^* N(F)$. These functors and this map are defined before any equivalence is asserted. On E° use coordinate $u = x/t$; thus q_∞ is $u = 0$ and q_0 is $v = u^{-1} = 0$.

Lemma 9.1 (The constant nearby calculation and its maps). *On the smooth locus of X'_s , $R\Psi_{X'} Q = Q$. At a node with chart $zw = t$, geometric nearby cohomology is Q in degree zero and $Q(-1)$ in degree one, generated by the Kummer class of z ; there is no other cohomology. At the old point of $xy = t^2$, the same statement holds with generator the Kummer class of x . These identifications are compatible with Kummer classes of units on smooth open neighbourhoods.*

Proof. The chart for a new node is $\mathbf{N} \rightarrow \mathbf{N}^2$ with $1 \mapsto (1, 1)$. For the old node it is $\mathbf{N} \rightarrow P$ with $1 \mapsto c$. Its group quotient is freely generated by a modulo c , and its underlying scheme is exactly the toric base change. Thus the logarithmic chart criterion gives a vertical fs log smooth morphism in both cases. In particular the relation $a + b = 2c$ does not introduce torsion in $P^{\text{gp}}/\mathbf{Z}c$.

Apply Nakayama's constant log-local-acyclicity theorem and the classical comparison, in [3, Theorem 8.3, Corollary 8.4]. The Kummer calculation of the resulting geometric log fibre gives exterior powers of the relative rank-one characteristic group, with its Tate twist ([3, §5]). The degree-one map is the actual Kummer boundary of the indicated monomial. At a smooth point the relative group is zero and smooth local acyclicity gives the degree-zero unit. These assertions may first be made with $\mathbf{Z}/\ell^r$; the displayed cohomology groups are free and compatible with reduction, so the adic constant computation gives the asserted rational result. The Kummer maps themselves are functorial, rather than chosen isomorphisms of the resulting vector spaces. For a unit h on a smooth neighbourhood its finite torsors $z^{\ell^r} = h$ are étale there; smooth specialization identifies their Kummer boundary with that of the specialized unit. This verifies the compatibility used below. $\square$

Theorem 9.2 (Intrinsic admissibility of the actual nearby image). *The functor $\mathcal{S}$ is an equivalence $\mathcal{V} \simeq \mathcal{U}^b(E^\circ)$, strong monoidal on this source. For every $F \in \mathcal{V}$ the actual attaching map $\alpha(F)$ is an equivalence, and the actual unit gives*

$$N(F) \xrightarrow{\sim} Rj_* \mathcal{S}(F).$$

Consequently N identifies $\mathcal{V}$ with the intrinsically defined category $\mathcal{C}_ = \{H \in \mathcal{C}(E, \mathbf{Z}) : i^! H = 0\}$.*

Proof. First take $F = Q$. The smooth neighbourhood of E° is $\text{Spec } R[u, u^{-1}]$, so $\mathcal{S}(Q) = Q_{E^\circ}$ by the actual smooth specialization unit. At q_0 , the source of $\alpha_a(Q)$ has its degree-one generator represented by $v = t/x$ on the geometric generic fibre. This class restricts, under the same specialization construction, to the Kummer class of the unit v on the punctured strict neighbourhood of q_0 in E . The latter generates $H^1(B_a)$. At q_∞ the identical argument uses $u = x/t$.

More explicitly, apply the exact nearby functor to the Kummer morphism $Q \rightarrow Q(1)[1]$ on the generic fibre, and precompose with its constant specialization unit. Naturality of

the open unit defining α identifies the image with the Kummer morphism on the smooth open stratum. The finite unit torsors there extend étale over $R[u, u^{-1}]$ or $R[v, v^{-1}]$, so the last assertion of Lemma 9.1 identifies this morphism before taking cohomology. The residue calculation of Proposition 5.1 then shows that its degree-one map is an isomorphism. In the common coordinate x , the signs are negative at q_0 and positive at q_∞ : $[x] = -[v]$ and $[x] = [u]$ after the geometric base class $[t]$ vanishes. Degree zero is the identity by the same unit, and there are no other degrees. Thus $\alpha(Q)$ is an equivalence.

Now use the already defined natural transformation $N \rightarrow Rj_*\mathcal{S}$. Its equivalence locus is thick. Both its values and those of $\mathcal{S}$ on Q are bounded in the required categories by the constant computation; exactness and retracts give the same property on $\mathcal{V} = \text{thick}(Q)$. This proves the comparison for every coefficient without assuming a lisse comparison theorem for those coefficients. Membership in $\mathcal{C}_*$ follows from Theorem 6.2.

It remains to check that $\mathcal{S}$ is an equivalence, rather than just a functor into that stratum category. Its map on the endomorphism algebra of Q is the actual specialization map

$$R\Gamma(\mathbf{G}_{m, \bar{K}}, Q) \longrightarrow R\Gamma(E^\circ, Q).$$

It is the identity in degree zero and sends the Kummer class of x to that of u . Indeed $x = tu$ and compatible roots of t in $\bar{K}$ identify the generic torsors with those of u ; the latter extend over the smooth neighbourhood. Both algebras have cohomology only in degrees zero and one, so this actual algebra map is an equivalence. Unit generation on both sides proves full faithfulness and essential surjectivity. Finally nearby cycles have their natural lax monoidal map; on E° its unit map is an equivalence by smooth local acyclicity. The tensor comparison is an equivalence on (Q, Q) , then on all pairs in the two thick closures by exactness in each variable. This proves the stated strong monoidal property of $\mathcal{S}$, not of N for the ambient tensor on E . $\square$

Proposition 9.3 (The tensor product appropriate to the star image). *Let $H = Rj_*Q_U$ with its actual commutative algebra structure on E . The forgetful functor from the thick subcategory generated by H in H -module sheaves to ordinary sheaf complexes is an equivalence onto $\mathcal{C}_*$. Thus the latter has a canonical tensor product $\otimes_H$ realized by module sheaves, with unit H , and*

$$N(F) \otimes_H N(G) \xrightarrow{\sim} N(F \otimes_Q G).$$

For a supported coefficient one likewise has

$$N(F) \otimes_H j_!\mathcal{S}(G) \simeq j_!(\mathcal{S}(F) \otimes_Q \mathcal{S}(G)).$$

These statements use a relative tensor, not the non-closed ordinary Q_E -tensor product exhibited earlier.

Proof. The algebra H is defined by the lax monoidal right adjoint Rj_* , and $j^*H = Q_U$. For the module generator,

$$R\text{Hom}_{H\text{-Mod}(E)}(H, H) = R\Gamma(E, H) \simeq R\text{Hom}_E(H, H).$$

The last equivalence is induced by $Q_E \rightarrow H$. Since its open restriction is the identity, its cone is

$$i_* \text{cofib}(Q_Z \longrightarrow L(Q_U)) \simeq \bigoplus_{a \in \mathbb{Z}} i_{a,*} Q(-1)[-1].$$

Indeed applying i^* gives exactly the displayed endpoint cofibres. For any closed coefficient C , adjunction gives $R\text{Hom}_E(i_*C, H) = R\text{Hom}_Z(C, i^!H) = 0$. Applying this to the actual cone proves the required equivalence. The larger object $i_*L(Q_U)$ is instead the cone of

$j_!Q_U \rightarrow Rj_*Q_U$. The forgetful map on generator endomorphisms is exactly this equivalence, since evaluation of an H -linear endomorphism at its unit gives the same section. The two-variable thick-generation argument proves full faithfulness on the thick module subcategory. Its image is $\text{thick}_E(H) = \mathcal{C}_*$, by the already proved open reconstruction. This supplies the claimed module realization without merely transporting an abstract tensor across an equivalence.

The module tensor is exact in each variable and takes (H, H) to H , so it preserves this thick subcategory. The actual lax tensor comparison for Rj_* factors over H and is an equivalence on (Q_U, Q_U) . Generation in each variable proves it for all unipotent coefficients. The comparison $N \rightarrow Rj_*\mathcal{S}$ is the open-adjunction unit and respects these module actions; together with the proved monoidality of $\mathcal{S}$ this gives the first formula. Finally the relative open projection formula, with $j^*H = Q_U$, gives the second formula and its actual H -action on the supported object. No unrestricted ordinary tensor closure has been added as an axiom. $\square$

9.1 The actual complete-support partner

Theorem 9.4 (Exceptional support on the modified model). *For $F \in \mathcal{V}$, the actual exceptional restriction satisfies*

$$\kappa^!R\Psi_{X'}F \xrightarrow{\sim} j_!\mathcal{S}(F).$$

This identifies the supported functor with $\mathcal{C}_!$ and exchanges it with the star functor of Theorem 9.2 under geometric Verdier duality. The comparisons are those induced by nearby duality and the actual open-immersion adjunctions.

Proof. Use the geometric nearby-duality theorem of Gabber in the formulation of [10, Corollary 3.2, Remark 3.3], with dualizing complexes relative to the geometric generic and special points. It gives $D_{X'_s}R\Psi F = R\Psi D_{X'_\eta}F$. The model is separated of finite type over the complete excellent trait, and the source consists of bounded constructible lisse perfect complexes. The cited construction is induced by evaluation and the six-operation trace, so it supplies the comparison of pairings as well as of objects. We use its geometric version; no assertion about taking wild-inertia invariants or about a full sliced vanishing-cycle functor is needed. For the present rational source it suffices to establish this canonical comparison on Q and its thick closure. On Q , the semistable constant computations have uniformly bounded finite free cohomology at every coefficient precision. The finite-coefficient nearby-duality maps are compatible with coefficient reduction and evaluation, giving the adic and then rational comparison in the constructible formalism. The equivalence locus of the two exact contravariant functors is closed under finite cones and retracts, so Proposition 4.4 extends it to $\mathcal{V}$. This argument does not exchange duality with an arbitrary ambient inverse limit of sheaves.

Write $D_\eta F = F^\vee(1)[2]$ on the smooth generic curve. Proper closed-immersion Verdier exchange and the preceding theorem give

$$\kappa^!R\Psi F \simeq D_E \kappa^* R\Psi D_\eta F \simeq D_E Rj_* \mathcal{S}(D_\eta F) \simeq j_!\mathcal{S}(F).$$

For the last step, strong monoidality preserves lisse duals and their evaluation, while exactness preserves shifts. On the smooth neighbourhood $\text{Spec } R[u, u^{-1}]$ of E° , the finite étale sheaves μ_{ℓ^r} specialize by their actual smooth pullbacks. These identifications commute with $\mu_{\ell^{r+1}} \rightarrow \mu_{\ell^r}$ and the coefficient evaluation maps. The selected adic passage and rationalization therefore give $\mathcal{S}(Q(1)) \simeq Q(1)$ canonically, without choosing a basis of the Tate line. Together these identifications give $\mathcal{S}(D_\eta F) \simeq D_{E^\circ} \mathcal{S}(F)$, compatibly with evaluation. All arrows come from these adjunctions and their Verdier mates, and therefore retain the evaluation and localization compatibilities. The support on the left is $E \hookrightarrow X'_s$, not the endpoint support $Z \hookrightarrow E$ used to prove the intrinsic gluing theorem. $\square$

Theorem 9.5 (One modification: star modules and supported values). *Let $i_Y : Y \hookrightarrow X_s$ and put*

$$B(F) = i_Y^* R\Psi_X F, \quad A_Y = B(Q), \quad B_!(F) = i_Y^! R\Psi_X F.$$

The actual cochain algebra map $A_\eta = R\Gamma(\mathbf{G}_{m,\bar{K}}, Q) \rightarrow A_Y$ is an equivalence. The functor B , with its natural A_Y -module structure, identifies $\mathcal{V}$ with $\text{Perf}(A_Y)$. For this actual modification the proper nearby and proper base-change maps give

$$B(F) \xrightarrow{\sim} R\Gamma(E, N(F)), \quad B_!(F) \xrightarrow{\sim} R\Gamma(E, \kappa^! R\Psi_X F) \simeq R\Gamma_c(E^\circ, \mathcal{S}(F)).$$

Together with Theorems 9.2 and 9.4, these give a Morita equivalence $\text{Perf}(A_Y) \simeq \mathcal{C}_$ with its actual supported partner. They do not give an equivalence with the full $\mathcal{C}(E, Z)$.*

Proof. The restriction map $R\Gamma(X_{\bar{\eta}}, F) \rightarrow B(F)$ is constructed by geometric nearby specialization. On Q it preserves the unit and sends $[x]$ to the primitive relative Kummer class at the old node, by Lemma 9.1. It is therefore an algebra equivalence; no factor of two occurs since a generates $P^{\text{gp}}/\mathbf{Z}c$. Exactness and unit generation extend this actual comparison to F , with its cup-product module action. The usual Morita identification for $\mathcal{V}$ then proves the first assertion. Forgetting this module structure would not give a fully faithful functor into $D(Q)$.

The map μ is proper and $\mu_{\bar{\eta}}$ is the identity. Classical proper nearby base change gives $R\Psi_X F \simeq R\mu_{s,*} R\Psi_{X'} F$. Proper base change along i_Y identifies its ordinary restriction with the first displayed integral on the complete inverse image E . Taking its Verdier mate gives the exceptional proper base-change comparison and hence the second integral. Inserting Theorem 9.4 and using properness of E gives the compact-support expression. Naturality of nearby restriction, cup products, the proper trace and their adjunctions identifies these with the same coefficient maps already used in the Morita comparison.

On $\mathcal{C}_*$, $R\Gamma(E, -)$ with its cochain action is the Morita functor for the generator $Rj_* Q$: the map $Q_E \rightarrow Rj_* Q$ induces $R\text{Hom}_E(Rj_* Q, H) \simeq R\text{Hom}_E(Q_E, H)$ because $i^! H = 0$. Thus the integral comparison here retains a full module-valued reconstruction on the proved admissible subcategory. This explains why it avoids the disjoint-skyscraper obstruction rather than contradicting it. $\square$

Corollary 9.6 (Supplied compatible finite descent). *Suppose a finite group Δ is supplied with compatible R -automorphisms of X and X' , preserving Y and E , commuting with μ , and preserving the chosen geometric base. The comparisons in Theorems 9.2–9.5 then induce equivalences with this actual finite descent retained. The Morita generators are induced generators as in Proposition 4.6, rather than only equivariant trivial coefficients.*

Proof. Generic restriction, geometric nearby cycles, the immersions of the invariant strata, and proper direct image form equivariant functors with their geometric composition cells. All comparison maps used above are their units, base-change maps or Verdier mates. They therefore define equivariant natural equivalences, whose limits over $B\Delta$ are equivalences of the actual descent categories. The source and target geometric conditions are invariant under pullback, so Proposition 4.6 supplies the stated generators. This argument neither exchanges a root-level colimit with $B\Delta$ nor constructs a lift of an arbitrary group action to the modification. $\square$

The supported value is equivalently $D_Q B(F^\vee)(-1)[-2]$, by the actual duality comparisons above. For the Jordan coefficient of Example 5.3 it has one one-dimensional group in each of degrees one and two, paired with the kernel and coinvariant groups of the dual local system. This checks the supported shift without changing the coefficient to a constant one.

The scope is one fixed modification and geometric rational unipotent coefficients. The two endpoint attaching maps, their Verdier partners and the complete inverse-image support have now been compared. The next section extends the pairwise comparisons to the stated supports and minimal kernels in the fixed-thickness integer-cut family. The geometry supplies common refinements; simultaneous higher profile coherence remains a separate target. Arbitrary model refinements, retained base Galois actions, unsupplied descent lifts and comparison with the absolute logarithmic finite-coefficient theorem require further results. The ordinary tensor obstruction for $\mathcal{C}_*$ remains valid: the stratum functor $\mathcal{S}$ is strong monoidal, and the actual star image uses the separately verified relative tensor $\otimes_H$ of Proposition 9.3.

10 Cellular refinement for a fixed thickness

10.1 The local input for every integer thickness

Proposition 10.1 (All-thickness nearby algebras and branch maps). *For every $d \geq 1$, give $T_d = \operatorname{Spec} R[z, w]/(zw - t^d)$ the chart $P_d = \langle a, b, c : a + b = dc \rangle$, with $a \mapsto z$, $b \mapsto w$ and $c \mapsto t$. Over the standard log trait this is a vertical fs log smooth scheme. For $\Lambda_r = \mathbf{Z}/\ell^r$, its geometric constant nearby stalk at the node has cohomology Λ_r and $\Lambda_r(-1)$ in degrees zero and one, and zero otherwise. The class $[z]$ is primitive. The actual open-adjunction map to the link on the branch with punctured coordinate z sends it to $[z]$; on the other branch $[w] = -[z]$ maps to its primitive coordinate class. These are equivalences of the actual cochain algebras. The same statements hold adically and over Q , including when p divides d .*

Proof. In coordinates $a = (1, 0)$, $c = (0, 1)$, $b = (-1, d)$, the monoid is exactly the integral cone $n \geq 0$, $dh + n \geq 0$: for $h < 0$ write $(h, n) = (-h)b + (n + dh)c$, and for $h \geq 0$ use $ha + nc$. It is fine and saturated, its group is $\mathbf{Z}a \oplus \mathbf{Z}c$, and the group map $\mathbf{Z} \rightarrow P_d^{\text{gp}}$ has zero kernel and free cokernel $\mathbf{Z}a$. The scheme map to the chart base change $\operatorname{Spec} R \times_{\operatorname{Spec} \mathbf{Z}[\mathbf{N}]} \operatorname{Spec} \mathbf{Z}[P_d]$ is an isomorphism. These verify the log smooth chart criterion, without an invertibility condition on d . Inverting c makes both a and b invertible, so the structure is vertical; the generic fibre has trivial log structure.

Apply [3, Theorem 8.3 and Corollary 8.4] with invertible finite coefficients. On the geometric log fibre the relative characteristic group is $P_d^{\text{gp}}/\mathbf{Z}c = \mathbf{Z}a$. Its prime-to- p Kummer inertia has the rank-one exterior cohomology computed in [3, §5]. The resulting degree-one class is the Kummer boundary of z , not $d[z]$. This proves the stalk assertion; on either smooth branch open the relative group is zero and smooth local acyclicity gives Λ_r .

To identify the map, apply nearby cycles to the generic Kummer morphism $\Lambda_r \rightarrow \Lambda_r(1)[1]$ of z and precompose with the specialization unit. On $D(z) = \operatorname{Spec} R[z, z^{-1}]$ the torsor $h^{\ell^r} = z$ is étale. Smooth specialization and naturality of the open-adjunction unit identify its boundary on the punctured strict branch with $[z]$. This computes the actual map in degree one; in degree zero it preserves the unit. The identical argument on $D(w)$ uses w . On the geometric generic fibre $[t] = 0$, hence $[w] = d[t] - [z] = -[z]$. Both source and target have only the two stated cohomology groups, so the algebra maps are equivalences. No formality of either algebra is used.

This computation is functorial in the monomial maps of the models. In particular $x = t^a z = t^b/w$ gives $[x] = [z] = -[w]$ after passage to the geometric generic fibre. A change by a smooth unit is handled by the same étale torsor naturality, with its actual specialized unit class retained. Pullbacks under the cut contractions below are these monomial maps; their proper direct-image units will be compared in Lemma 10.9. We do not infer a shriek comparison from this stalk computation. Finally $p \mid d$ introduces no relative group torsion and changes none of these arguments: only ℓ is required to be invertible.

For the coefficient passage, the finite-level Kummer maps commute with reduction. At nodes their cohomology is finite free in the fixed range $[0, 1]$, and on smooth strata in degree zero. Thus the actual systems have derived reduction equal to the lower-precision system; their stalk cohomology towers are Mittag–Leffler. In the constructible adic formalism their reductions detect equivalences. The compatible finite-level algebra comparisons consequently give an adic equivalence and then its rationalization. This passage uses the constructed constant system, not inverse limits of arbitrary constructible sheaves. $\square$

Lemma 10.2 (Naturality of the selected branch maps). *The branch maps in Proposition 10.1 are maps of cochain algebras obtained by open adjunction. Their coefficient versions are module maps. For the monomial model maps of Proposition 10.3, they give the open-unit squares and the two-step attachment comparison of Lemma 10.10, with both branch signs retained.*

Proof. At precision r , define every map by applying the open-adjunction unit to the nearby complex, then restricting to the chosen branch. For a refinement u , let F_u denote module pullback and let $\chi_u : F_u Rj_* \rightarrow Rj'_* F_U$ be adjoint to the open counit. Thus the actual square is

$$\begin{array}{ccc} F_u N_I(F) & \longrightarrow & N_J(F) \\ \downarrow & & \downarrow \\ F_u Rj_* j'^* N_I(F) & \xrightarrow{\chi_u} & Rj'_* j'^* N_J(F). \end{array}$$

The upper map is the nearby mate constructed in Lemma 10.9. The bottom arrow includes Rj'_* of the open restriction of that upper comparison, after χ_u . To check the square, apply open restriction and take the adjoint: both composites become the same open counit followed by the same nearby pullback. Restricting this square to a closed fibre gives the attachment square, without assuming the external link is cartesian on that fibre.

On a gap $[a, b]$, the coordinate equalities $x = t^a z = t^b/w$ give, on the geometric generic fibre,

$$[x] = [z], \quad [w] = -[x].$$

For a refined gap $[a', b'] \subset [a, b]$, pullback sends z to $t^{a'-a} z'$ and w to $t^{b-b'} w'$. Naturality of the Kummer connecting morphism therefore gives the same two classes, with the same signs. This checks both branches of the actual square, not just the rank of its degree-one groups. Multiplication by a smooth unit contributes its own specialized Kummer class; it is not silently discarded. Base constants disappear here only after geometric generic base change.

For $I \subset J \subset K$, the exponents add and the mate of $\chi_v F_v \chi_u$ is the composite open counit, which is the mate of χ_{uv} . Their canonical adjunction comparison therefore gives $\chi_{uv} = \chi_v F_v \chi_u$ with the specified composition cell. The closed restriction gives the two-step formula for the branch attachments, including the two branches at an internal node. All maps were defined before the Kummer computation. They are algebra maps at the unit by the finite-arity counit calculation in Lemma 10.9, and module maps on coefficients. The loci of equivalence for these exact natural transformations are thick. Unit generation therefore proves the assertions for nontrivial rational unipotent coefficients as well, without splitting their extensions. The coefficient passage is Lemma 4.1. This verifies these specified maps and their two-step composition, not the simultaneous higher coherence of all supported and kernel operations. $\square$

10.2 An actual family closed under common refinement

Fix an integer $m \geq 2$ and put $X_m = \operatorname{Spec} R[x, y]/(xy - t^m)$. The coefficient source remains $\mathcal{V} = \mathcal{U}^b(\mathbf{G}_{m, \bar{K}})$; geometric nearby cycles in this section forget the base Galois action. For a

set $I \subset \{1, \dots, m-1\}$ write

$$0 = a_0 < a_1 < \dots < a_r < a_{r+1} = m, \quad d_i = a_{i+1} - a_i.$$

Define X_I by gluing the charts

$$\text{Spec } R[z_i, w_i]/(z_i w_i - t^{d_i}), \quad z_i = x/t^{a_i}, \quad w_i = t^{a_{i+1}}/x,$$

with $w_{i-1} = z_i^{-1}$ on the consecutive principal-open overlaps. The induced nonconsecutive overlaps lie in the common generic fibre. Let $E_I = X_I \times_{X_m} Y$, where $Y = V(x, y, t)$. For $I = \emptyset$, this means $X_I = X_m$ and $E_I = Y$.

Proposition 10.3 (Models, complete support, and their maps). *These are actual proper toroidal modifications of X_m , with identical geometric generic fibres. For $I \neq \emptyset$, E_I is a reduced chain of r projective lines. Its strata are r copies U_i of $\mathbf{G}_m$ and the $r+1$ nodes q_i of the ambient special fibre; the two outside nodes are smooth points of the chain itself. For $I \subset J$ there is a canonical proper map $\mu_{JI} : X_J \rightarrow X_I$, and $\mu_{KI} = \mu_{JI}\mu_{KJ}$ for $I \subset J \subset K$. The complete inverse image of E_I is E_J . Common refinements are supplied by $I \cup J$.*

Proof. On a gap of thickness d , inserting a cut at e , $0 < e < d$, is the blow-up of (z, t^e) in $A = R[z, w]/(zw - t^d)$. Its Rees algebra is $A[zs, t^e s] \subset A[s]$; the two standard opens have rings $A[t^e/z]$ and $A[z/t^e]$ inside the fraction field. With $v = t^e/z$ and $u = z/t^e$, respectively, their fraction-ring charts are

$$R[z, v]/(zv - t^e), \quad w = t^{d-e}v, \quad R[u, w]/(uw - t^{d-e}), \quad z = t^e u.$$

There is no extra component from an uncanceled equation. The monoid $P_d = \langle a, b, c : a + b = dc \rangle$ is the saturated cone $\{(h, n) : n \geq 0, dh + n \geq 0\}$, with $a = (1, 0)$, $c = (0, 1)$, $b = (-1, d)$. The two blow-up monoids are already saturated copies of P_e and P_{d-e} . Thus this is also the actual fs log blow-up; no root extension of the base is required.

Iterating gives the displayed gap charts. If $[a', b'] \subset [a, b]$ is a refined gap, the map to the old chart is $z \mapsto t^{a'-a}z'$ and $w \mapsto t^{b-b'}w'$. These formulas agree on overlaps and compose identically. Each insertion is a blow-up and hence proper; the resulting glued models are independent of insertion order by these formulas. The chart formulas construct the identifications for different insertion orders. These identifications are unique once fixed on the dense generic fibre, since the sources are reduced and the targets are separated. The common generic fibre alone does not construct them.

On an interior gap the ideal of Y pulls back to (t) ; on the first and last gaps it pulls back to (t, z_0) and (t, w_r) , respectively. The reduced special fibres of the gap charts and their inverse coordinate overlaps therefore give precisely the stated chain. The empty cut set gives the point instead. Equality of complete inverse-image supports follows from composition of these ideal pullbacks. Finally the displayed formulas construct all maps to the model for $I \cup J$, proving the common-refinement assertion as an actual scheme statement. $\square$

The family is deliberately restricted to integer cuts of this fixed thickness. Gaps of length one are regular semistable charts; larger gaps are toroidal nodes of the stated thickness. This theorem neither inserts another regular node into a thickness-one chart over the same base nor treats arbitrary ramified base extensions.

10.3 Intrinsic cellular coefficients with branchwise admissibility

Put $\kappa_I : E_I \rightarrow (X_I)_s$ and define the actual commutative nearby algebra and complete-support module

$$H_I = \kappa_I^* R\Psi_{X_I} Q, \quad W_I = \kappa_I^! R\Psi_{X_I} Q.$$

On U_i the algebra H_I is Q . At a node write $A_q = i_q^* H_I$. For an incident branch U_i write B_{qi} for its punctured-strict- neighbourhood cochain algebra. The actual open-adjunction unit gives $\lambda_{qi} : A_q \rightarrow B_{qi}$.

Lemma 10.4 (The branch algebras and their coefficient functors). *Every A_q has cohomology Q and $Q(-1)$ in degrees zero and one. Both actual algebra maps*

$$A_q \xrightarrow{\lambda_{qi}} B_{qi} \xleftarrow{\kappa_{qi}} A_i = R\Gamma(U_i, Q)$$

are equivalences. The coefficient link is consequently an equivalence $\mathcal{U}^b(U_i) \rightarrow \text{Perf}(A_q)$, with the A_q -action obtained by restriction along λ_{qi} .

Proof. For every gap, $P_d^{\text{gp}}/\mathbf{Z}c$ is generated by a . Proposition 10.1 therefore supplies the all-gap calculation, with its primitive Kummer class and no factor of d . The branch coordinates are x/t^{a_i} and their inverses. The actual Kummer-morphism argument in Theorem 9.2 computes each λ_{qi} , independently on the two branches at an internal node. On the other side, κ_{qi} is the actual restriction from $\mathbf{G}_m$ to a punctured strict neighbourhood, as in Proposition 5.1. Their naturality as actual maps, including coefficient extensions and the two branch signs, is Lemma 10.2. Both preserve the unit and take the primitive degree-one class to a generator, with the sign of the local coordinate. They are therefore algebra equivalences. The coefficient link, as a module-valued functor, is extension of scalars along the second map followed by restriction along the first; the specified evaluation map is

$$R\Gamma(U_i, V_i) \otimes_{A_i}^L B_{qi} \longrightarrow L_{qi}(V_i).$$

It is B_{qi} -linear, and then A_q -linear by restriction along $A_q \rightarrow B_{qi}$. It is an equivalence at the unit and hence on its thick closure by exactness. Thus it is an equivalence of the stated categories, not just a comparison of underlying cohomology complexes. $\square$

Let $S \subset E_I$ be a locally closed union of the stated strata. Write $H_S = H_I|_S$ and use actual H_S -module sheaves. Empty supports are assigned the zero category. The incidence graph of a nonempty S has a vertex for each included U_i and q , and an edge for each included branch incidence; it is a finite forest.

Definition 10.5 (The cellular coefficient category). Let $\mathcal{C}_I(S)$ consist of bounded constructible H_S -modules with the following geometric conditions. Their restriction V_i to U_i has unipotent lisse cohomology. Their node restriction M_q is perfect over A_q . For every included branch the component

$$M_q \longrightarrow L_{qi}(V_i)$$

of the actual attaching map is an equivalence of A_q -modules. The module actions and these arrows are part of the actual sheaf module structure; no condition refers to an image from the old model.

At an internal node with two branches it is the two individual maps that are invertible. The combined map $M_q \rightarrow L_{q,i}(V_i) \oplus L_{q,i+1}(V_{i+1})$ is diagonal up to those identifications, and need not be invertible. Thus the one-component criterion $i^! F = 0$ must not be imposed on the whole chain. Nor do we forget the A_q -actions and demand full faithfulness in the category of ordinary node complexes. Branchwise admissibility is a geometric condition to be checked on the actual maps; perfectness of an integrated cochain module alone does not imply it.

Theorem 10.6 (Cellular reconstruction before invariance). *The category $\mathcal{C}_I(S)$ is the homotopy limit of its forest of stratum categories and the actual branch link equivalences of Lemma 10.4. It is the thick subcategory generated by H_S in H_S -module sheaves. Hence*

$$\mathcal{C}_I(S) \simeq \text{Perf}(R\Gamma(S, H_S)).$$

For $S = E_I$, the actual functor $N_I(F) = \kappa_I^* R\Psi_{X_I} F$, with its H_I -action, is an equivalence $\mathcal{V} \simeq \mathcal{C}_I(E_I)$.

Proof. Use the open union of the included U_i and its finite closed complement of nodes in S . Recollement in module sheaves reconstructs an object as

$$Rj_* V \underset{i_* L(V)}{\times}^h i_* M.$$

Here $j : U \rightarrow S$ is the included open-stratum union, $i : Z \rightarrow S$ its closed node complement and $L = i^* Rj_*$. The H_S -action on $Rj_* V$ uses $H_S \rightarrow Rj_* j^* H_S$, and the closed module uses $H_S \rightarrow i_* i^* H_S$. Forgetting the module action creates limits and detects equivalences, so the underlying pullback can be checked on sheaf complexes. Its mapping spectra, however, are module mapping spectra, not arbitrary Q -linear mapping spectra. The selected open coefficients have bounded constructible links by unit generation and Lemma 10.4. Each perfect A_q -module has bounded finite-dimensional underlying Q -cohomology, since A_q does and finite cones and retracts preserve that property. The finite pullback therefore stays in the stated bounded constructible category. No unbounded-module assertion is used. The generator endomorphism identification has type

$$R\text{Hom}_{H_S\text{-Mod}(S)}(H_S, H_S) \simeq R\Gamma(S, H_S).$$

At a node the punctured strict neighbourhood is the disjoint union of its included branches, so $L(V)$ is their finite product. The module mapping spectrum is the fibre of the same difference map as in Theorem 6.2, now with A_q -linear maps at each node. These statements follow from the actual adjunctions and prove enhanced reconstruction, including the homotopies of attaching squares.

Imposing the individual equivalences in Definition 10.5 therefore gives exactly the asserted homotopy limit of categories. Removing a leaf of a tree forgets an object determined, with a contractible space of choices, by an equivalence across its single incident edge. Evaluation at any vertex of a connected tree is consequently an equivalence of stable categories. For a forest the result is a finite product of these categories. By Lemma 10.4, H_S evaluates to a generator at each vertex. Its finite product components split by the component idempotents, so it generates the entire forest category. Conversely H_S satisfies all the geometric conditions, and those conditions are stable under finite cones and retracts. This proves the thick identification and then the Morita assertion. The empty and isolated- node cases have the same statement with zero or one vertex.

For actual $N_I(F)$, each branch admissibility map is a natural transformation of exact functors on $\mathcal{V}$. Its value on Q is λ_{qi} , already proved invertible, and its node module on Q is A_q . Unit generation proves the conditions for every F . At any smooth component stratum the actual specialization functor is an equivalence by the coordinate calculation $[x] \mapsto [x/t^{a_i}]$ of Theorem 9.2. Evaluation on that vertex and the tree reconstruction prove the last assertion. With no cuts use the old-node module calculation with P_m instead. Thus the geometric admissibility criterion was verified before identifying the image; the image was not used as its definition. $\square$

Example 10.7 (An internal node is not a terminal node). Let q be an internal node of a chain with at least two components. After identifying both branch algebras with A_q by their actual Kummer maps, the attaching map of H_I is

$$A_q \xrightarrow{(1,1)} A_q \oplus A_q.$$

The localization triangle on the chain therefore gives

$$i_q^! H_I \simeq A_q[-1],$$

where an ordering of the branches identifies the quotient map with $(b_1, b_2) \mapsto b_2 - b_1$. At an outside node only one branch is present, so the same costalk is zero. These assertions follow from the actual link and localization triangle; the internal boundary is nonzero, including for nonconstant coefficient modules.

In particular the complete-support module is not generally extension by zero from the disjoint smooth strata. Nearby Verdier duality gives at an internal node

$$i_q^* W_I \simeq D_Q(A_q[-1])(-1)[-2] = D_Q A_q(-1)[-1],$$

which has nonzero cohomology in degrees zero and one. At an outside node its stalk is zero. Thus the one-component formula $W = j_! Q$ cannot be copied onto a longer chain; the cellular supported module retains the internal gluing.

10.4 Support categories and the data to be compared

We distinguish internal module duality from geometric Verdier duality:

$$M^{\vee_{H_S}} = R\mathcal{H}om_{H_S}(M, H_S), \quad D_S M = R\mathcal{H}om_Q(M, \omega_S), \quad \tilde{D}_S M = D_S M(-1)[-2].$$

Here $\mathcal{H}om$ denotes the sheaf internal Hom; its global sections are taken only when explicitly written. Module duality is an anti-equivalence on the finite-cell category $\mathcal{C}_I(S)$. Verdier duality uses the geometric dualizing complex ω_S , not H_S , and gives the anti-equivalence $\tilde{D}_S : \mathcal{C}_I(S)^{\text{op}} \rightarrow \mathcal{C}_I^!(S)$, with the inverse obtained by the same normalized duality. The same convention applies to the two ambient types. The action on a dual is the contragredient action, identifying the opposite algebra through commutativity. The normalization is involutive because duality reverses both Tate twists and shifts. The covariant expression $\tilde{D}_S(M^{\vee_{H_S}})$ composes two contravariant functors; it does not identify these two dualities.

For $k_S : S \hookrightarrow E_I$ define

$$W_S = k_S^! W_I = (\kappa_I k_S)^! R\Psi_{X_I} Q.$$

This is an actual support object. Geometric nearby duality identifies it, with its H_S -action, as

$$W_S \simeq D_S H_S(-1)[-2].$$

Let $\tilde{D}_S = D_S(-1)[-2]$; this is an involutive duality on constructible H_S -modules, using the commutativity of H_S to identify the opposite action. Let $\mathcal{C}_I^!(S)$ be its image on $\mathcal{C}_I(S)$, equivalently the modules whose normalized Verdier dual satisfies Definition 10.5.

Proposition 10.8 (Actual supported partners and typed operations). *Both categories are realized in actual module sheaves, and*

$$M \mapsto W_S \otimes_{H_S}^L M \simeq \tilde{D}_S(M^{\vee_{H_S}})$$

is an equivalence $\mathcal{C}_I(S) \simeq \mathcal{C}_I^!(S)$. For $S = E_I$ and $M = N_I(F)$ its value is $\kappa_I^! R\Psi_{X_I} F$. For an inclusion of allowed supports, ordinary restriction preserves $\mathcal{C}_I$, and exceptional restriction preserves $\mathcal{C}_I^!$. Tensor over H_S , its action on the supported partner, and their evaluation maps are preserved on these categories.

Proof. The formula for W_S is Gabber nearby duality followed by actual Verdier exchange for $\kappa_I k_S$, with the generic curve shift and Tate twist. It applies to every gap just as in Theorem 9.4. Thus it is a geometric identification of the already defined costalk. For a finite-cell H_S -module, the evaluation comparison from $W_S \otimes_{H_S} M$ to $\tilde{D}_S(M^{\vee_{H_S}})$ is an equivalence on H_S and hence on its thick closure. The right side is a composite of two dualities, so gives the asserted equivalence. For actual nearby coefficients the natural mixed projection and nearby cup-product map gives $W_I \otimes_{H_I} N_I(F) \rightarrow \kappa_I^! R\Psi F$. It is balanced for the actual H_I -action, is the identity at $F = Q$, and hence is an equivalence throughout the geometrically generated source. No unrestricted exceptional projection formula is assumed to be invertible on arbitrary constructible coefficients.

Restriction takes H_T to H_S , so the proved generation theorem shows it preserves the star categories. Verdier exchange gives the exceptional assertion for their supported partners. Relative tensor and evaluation preserve finite-cell modules by exactness and the unit calculation. The same argument proves the mixed action on W_S . $\square$

We now specify the *cellular datum* $\mathfrak{C}_I$. It consists of the categories $\mathcal{C}_I(S), \mathcal{C}_I^!(S)$ for the allowed supports, their actual algebras H_S , supported modules W_S and Morita presentations, restriction and exceptional restriction with their composition cells, and the actual extension/localization diagrams in the ambient module categories. In particular, for an open/closed decomposition $T = U \sqcup Z$, it retains

$$j_! j^* M \longrightarrow M \longrightarrow i_* i^* M \longrightarrow, \quad i_* i^! P \longrightarrow P \longrightarrow Rj_* j^* P \longrightarrow,$$

where $M \in \mathcal{C}_I(T)$ and $P \in \mathcal{C}_I^!(T)$. Extension terms are not asserted to belong to $\mathcal{C}_I(T)$: they have their own support type and live in the displayed ambient module category. The datum includes their mapping objects through recollement, the branch inertia algebras and maps, and the Verdier exchange of these triangles. Thus an invariance assertion for this datum includes the restriction/link functors and boundary maps, not only the value of $R\Gamma(E_I, -)$.

The ambient category needed here can itself be specified without using all constructible modules. Let $\mathcal{A}_I(S)$ have the same stratum conditions as Definition 10.5, but allow arbitrary A_q -linear attaching maps. Recollement proves that it is generated by $j_{i,!} Q_{U_i}$ and $i_{q,*} A_q$. It contains the displayed star localization terms; its normalized Verdier dual contains the dual terms. All their mapping spectra are the actual module mapping spectra. The following fibrewise statement ensures that these maps, including off-diagonal extension maps, are also compared under refinement. The roles can be summarized as

$$\mathcal{C}_I(S) \subset \mathcal{A}_I(S), \quad \mathcal{C}_J(S') \subset \mathcal{A}_{J|I}(S') \subset \mathcal{A}_J(S').$$

The first category requires every included branch map to be invertible; the middle fine category requires this only within coarse fibres; the ambient categories permit arbitrary attachments. The cartesian categories carry the coefficient equivalence, while the middle ambient category retains the coarse localization maps.

10.5 The proper cellular comparison

Let $I \subset J$, and put $u : E_J \rightarrow E_I$. For an allowed support $S \subset E_I$ use its *complete* inverse image $S' = u^{-1}S$, with the pulled-back scheme structure. Its strata have the preceding forest description; nilpotent structure, if present, does not change the etale module-sheaf category. The map $u_S : S' \rightarrow S$ is proper.

Lemma 10.9 (The actual algebra unit under contraction). *The nearby pullback map and its adjoint give*

$$u_S^* H_S \longrightarrow H_{S'}, \quad H_S \xrightarrow{\sim} Ru_{S,*} H_{S'}.$$

The second is an equivalence of actual algebras, natural for restrictions of supports and for composition of the model maps. There is also a natural equivalence $W_S \simeq Ru_{S,*}W_{S'}$ with its H_S -action.

Proof. We construct the map at each finite precision, then apply Lemma 4.1. Write $f : X_J \rightarrow X_I$, and let j_I, j_J denote the geometric generic maps in the nearby construction. The special pullback of the mate

$$f^* Rj_{I,*} F \longrightarrow Rj_{J,*} f_\eta^* F$$

is the nearby pullback map. The mate itself is adjoint to

$$j_J^* f^* Rj_{I,*} F \simeq f_\eta^* j_I^* Rj_{I,*} F \xrightarrow{f_\eta^* \epsilon} f_\eta^* F.$$

These formulas can be made before geometric generic base change and are compatible with it. For any finite tuple of coefficients, tensoring these counits and then multiplying is the counit for the tensor of the tuple. Thus the mate commutes with every finite-arity lax tensor operation, including the nullary unit and substitution. At $F = \Lambda_r$ it is a map in commutative algebras, and for other F it is a map in modules over that algebra. This constructs the lift; it is not inferred from a cohomology isomorphism.

The adjoint $\theta : H_S \rightarrow Ru_{S,*}H_{S'}$ is the classical proper nearby comparison. Indeed composition of direct images gives $Rf_* Rj_{J,*} = Rj_{I,*} Rf_{\eta,*}$, and ordinary proper base change moves the special restriction past Rf_* . Here f_η is the identity. This derives the required proper nearby theorem from ordinary proper base change (SGA 4, Expose XII, Theorem 5.1), without an additional stack-nearby assertion. Base changing again to the complete inverse image of S gives θ , hence its underlying equivalence and its equivalence in commutative algebras.

Write $F_u = u_H^*$ and $G_u = Ru_*$. Their module projection map is specified by the adjoint of the evaluation map

$$F_u(M \otimes_{H_S} G_u P) \simeq F_u M \otimes_{H_{S'}} F_u G_u P \xrightarrow{1 \otimes \epsilon} F_u M \otimes_{H_{S'}} P.$$

Its adjoint is the H_S -linear projection map

$$\pi_{M,P} : M \otimes_{H_S} G_u P \longrightarrow G_u(F_u M \otimes_{H_{S'}} P).$$

Fix P in the domain of these operations. For $M = H_S$ this map is

$$G_u P \xrightarrow{\eta_{G_u P}} G_u F_u G_u P \xrightarrow{G_u \epsilon_P} G_u P = G_u P,$$

whose composite is the identity by the adjunction triangle identity. Thus $\pi_{M,P}$ is invertible on $\text{thick}(H_S)$ by exactness; the counit ϵ_P itself need not be invertible on an arbitrary ambient object. Taking $P = H_{S'}$ gives the comparison for $G_u F_u M$; taking $P = W_{S'}$ gives the selected supported-action comparison.

For the supported map first use the finite-level nearby duality pairing to identify $W_S = \tilde{D}_S H_S$. The proper duality map is adjoint to the pairing

$$G_u H_{S'} \otimes G_u \tilde{D}_{S'} H_{S'} \longrightarrow G_u(H_{S'} \otimes \tilde{D}_{S'} H_{S'}) \longrightarrow G_u \omega_{S'}(-1)[-2] \xrightarrow{\text{Tr}_u} \omega_S(-1)[-2].$$

It gives $G_u W_{S'} \simeq \tilde{D}_S G_u H_{S'} \simeq W_S$. Moving an element of H_S from one input of this pairing to the other gives the same map: this is associativity of the algebra map $u^* H_S \rightarrow H_{S'}$ and the evaluation action. Equivalently the two composites from $H_S \otimes G_u W_{S'}$ to W_S agree. This proves the H_S -linear mate and its compatibility with evaluation. Reduction and rationalization are those of Lemma 4.1, including the pairings.

For an open or closed support restriction the maps are formed from the same counits after base change. Applying its restriction to the displayed projection and pairing formulas gives their support squares. In particular the open-unit square, whose closed restriction is the attaching map, is a square of modules; its explicit mate is Equation (1). This proves the individual module and support comparisons used below. It does not by itself construct a single coherent diagram containing all six-operation and kernel variances at once. $\square$

Lemma 10.10 (Comparison of the ambient gluing maps). *Let $\mathcal{A}_{J|I}(S')$ be the full subcategory of $\mathcal{A}_J(S')$ whose restriction to the complete inverse image of each coarse stratum belongs to its intrinsic cartesian category of Definition 10.5. Then actual module pullback and direct image induce mutually inverse equivalences*

$$\mathcal{A}_I(S) \simeq \mathcal{A}_{J|I}(S').$$

In particular this compares the full mapping spectra of the extension and localization terms for coarse supports, not only their restrictions.

Proof. Adjunction and the fibre category. Work first in the ambient derived categories of module sheaves. The algebra map $u_S^* H_S \rightarrow H_{S'}$ gives the genuine adjunction

$$F = u_H^* = H_{S'} \otimes_{u_S^* H_S}^L u_S^*(-) \quad \dashv \quad G = Ru_{S,*},$$

where G restricts the action along $H_S \rightarrow Ru_{S,*} H_{S'}$. The unit is the sheaf pullback/direct-image unit followed by the algebra map; the counit is induced by the direct-image evaluation $u_S^* Ru_{S,*} P \rightarrow P$ and the $H_{S'}$ -action.

Over a coarse open stratum u is an isomorphism. Over a coarse node q write $u_q : Z_q \rightarrow q$ for its complete fibre. The restricted left adjoint is

$$F_q(M) = H_{Z_q} \otimes_{A_q}^L M.$$

The actual algebra unit gives $A_q \simeq Ru_{q,*} H_{Z_q}$. For $M \in \text{Perf}(A_q)$, projection on the free module and then finite cones and retracts proves that $M \rightarrow Ru_{q,*} F_q(M)$ is an equivalence. Hence F_q is fully faithful. Its thick image contains H_{Z_q} , which generates the intrinsic cartesian fibre category by Theorem 10.6. Thus F_q is an equivalence onto that category, with counit invertible there. This proves the fibre statement without any arbitrary attaching maps.

Transport of arbitrary attachments. Let $j : U \rightarrow S$ be the union of coarse open strata and $i : Z \rightarrow S$ the closed node union. Use j', i' for their complete inverse images. The map over U is the identity. Put

$$L(V) = i^* Rj_* V, \quad L'(V) = i'^* Rj'_* V, \quad F_Z = \prod_q F_q, \quad G_Z = Ru_{Z,*}.$$

Ordinary proper base change and composition of right adjoints give an actual natural equivalence

$$\lambda_V : G_Z L'(V) \xrightarrow{\sim} i^* G Rj'_* V \xrightarrow{\sim} i^* Rj_* V = L(V).$$

It uses the actual algebra map of Lemma 10.9. The object $L'(V)$ need not be cartesian inside the exceptional fibres: an external branch can contribute only at a terminal point. The adjunction $F_Z \dashv G_Z$ below is therefore used on the full ambient module categories, even though F_Z is an equivalence only on the specified fibre subcategories.

We verify which attachment is induced by actual pullback. Let $\nu_E : E \rightarrow Rj_* j^* E$ be the localization unit, and define

$$\chi_V : F Rj_* V \longrightarrow Rj'_* V$$

to be adjoint, under $j'^* \dashv Rj'_*$, to $j'^*FRj_*V \simeq j^*Rj_*V \rightarrow V$. The last arrow is the open counit. This defines χ on all objects and morphisms, independently of generation. Base change for module pullback identifies $i'^*F = F_Z i^*$; set $\kappa_V = i'^*\chi_V : F_Z L(V) \rightarrow L'(V)$. The two maps

$$FE \xrightarrow{F\nu_E} FRj_*V \xrightarrow{\chi_V} Rj'_*V, \quad FE \xrightarrow{\nu'_{FE}} Rj'_*j'^*FE = Rj'_*V$$

are the same: under the open adjunction both correspond to the identity of V . This is an equality of natural transformations via the adjunction equivalence of mapping spaces. Hence the actual pullback of (V, M, α) has attachment

$$\beta_{\text{geom}} = \kappa_V \circ F_Z \alpha.$$

We next identify κ , rather than assuming it is the required mate. Denote the ambient units by η and η_Z . Consider

$$Rj_*V \xrightarrow{\eta} GFRj_*V \xrightarrow{G\chi_V} GRj'_*V \simeq Rj_*V.$$

The right-adjoint composition $GRj'_* \simeq Rj_*$ is the one from the common open immersion; over U , both F and G are the identity. Thus the composite restricts to the identity of V . Full faithfulness of Rj_* identifies its mapping space with that of V , so the composite is the identity, with its canonical homotopy. Applying i^* , and the proper base-change square that carries η to η_Z , gives

$$\lambda_V \circ G_Z \kappa_V \circ \eta_{Z, L(V)} = \text{id}_{L(V)}.$$

The adjoint of κ_V is consequently λ_V^{-1} . Writing the adjunction counit as ϵ_Z proves the explicit identity

$$\kappa_V = \epsilon_{Z, L'(V)} \circ F_Z(\lambda_V^{-1}), \quad \beta_{\text{geom}} = \epsilon_{Z, L'(V)} \circ F_Z(\lambda_V^{-1}\alpha). \quad (1)$$

This construction does not require $L'(V)$ to lie in the cartesian fibre category. Conversely, the adjoint of a fine attachment, followed by λ , recovers α . The already proved fibre equivalence identifies its closed object with $F_Z M$. The triangle identities therefore give inverse assignments on attachments and on their full mapping spaces.

For clarity, the mapping spectrum between coarse triples is

$$\text{fib} \left(\text{Map}(V, V') \oplus \text{Map}(M, M') \xrightarrow{(f, g) \mapsto L(f)\alpha - \alpha'g} \text{Map}(M, L(V')) \right).$$

The fine expression has $F_Z M, F_Z M'$ and $L'(V')$ in place of the closed objects and link. Fibre full faithfulness identifies the second summand, and the ambient adjunction identifies

$$\text{Map}(F_Z M, L'(V')) \simeq \text{Map}(M, G_Z L'(V')) \simeq \text{Map}(M, L(V')).$$

Naturality of the mates identifies the difference maps. Thus the whole mapping spectra agree, including off-diagonal terms, and every fine triple in the stated category is reconstructed from a coarse one. This proves full faithfulness and essential surjectivity of actual F , not just of an abstract generator correspondence. Equivalently its actual unit can be checked on all coarse strata by proper base change: it is the identity on U and the already proved unit $M_q \rightarrow Ru_{q,*}F_q M_q$ on each q . The strata are jointly conservative on the underlying sheaf complexes.

Two refinements and a nontrivial fibre. For $S'' \xrightarrow{v} S' \xrightarrow{u} S$, use the same coarse open U and its complete inverse images. Write F_u, F_v for module pullback and keep the analogous closed restrictions. The definition by the open counit gives

$$\chi_{uv, V} = \chi_{v, V} \circ F_v \chi_{u, V}, \quad \kappa_{uv, V} = \kappa_{v, V} \circ F_{v, Z} \kappa_{u, V}$$

under $F_{uv} = F_v F_u$. Indeed both composites are adjoint to the identity over U . Likewise the right-adjoint composition gives $\lambda_{uv} = \lambda_u \circ G_{u,Z} \lambda_v$. Consequently an arbitrary attachment satisfies

$$\beta_{uv}(\alpha) = \kappa_{v,V} F_{v,Z} (\kappa_{u,V} F_{u,Z} \alpha) = \beta_v(\beta_u(\alpha)).$$

These comparison homotopies are obtained in the functor adjunctions, so for three refinements their associativity is that of composition and its mates. They are not choices made after an objectwise test.

For a concrete fibre with an internal node take thickness 6, $I = \{1, 5\}$, $J = \{1, 2, 4, 5\}$ and $K = \{1, 2, 3, 4, 5\}$. The coarse node between cuts 1 and 5 has in the J model a complete fibre containing the components at 2 and 4 and their internal node. Choose arbitrary open coefficients V on the two coarse branches, $M \in \text{Perf}(A_q)$, and an arbitrary $\alpha : M \rightarrow L(V)$; no invertibility or diagonal condition is imposed. Its fine closed coefficient is $H_{Z_q} \otimes_{A_q}^L M$, while $L'(V)$ is supported at the two external attachment points and vanishes on the fibre's interior, including the internal node. Equation (1) therefore transports both arbitrary external attachments without making this link cartesian. Adding cut 3 refines the internal node; the displayed composition identity proves that its attachment is exactly that of the direct $I \rightarrow K$ pullback. This tests the off-diagonal extension map, not just the coefficient H or its cohomology.

Extension terms and boundary cells. The standard objects have triples $(V, 0, 0)$ and $(0, M, 0)$, namely $j_! V$ and $i_* M$. Their pullbacks are $j_!^* V$ and $i_*^* F_Z M$. These generate the fibrewise-cartesian ambient category by the actual open/closed localization triangle and the fibre generation theorem; they are not replaced by the single global generator H_S . Their units are the open identity and the fibre unit, and their counits are the corresponding open identity and fibre counit. In particular

$$\text{Map}(i_* M, j_! V[1]) \simeq \text{Map}(M, L(V)) \simeq \text{Map}(F_Z M, L'(V)) \simeq \text{Map}(i_*^* F_Z M, j_!^* V[1]).$$

Using the same localization-triangle convention on both sides, its extension class α maps to β . Thus the actual boundary and not merely the two terms is compared. General coarse-support localization follows by restriction of these squares; Verdier mates give the dual triangles. No equivalence with all of $\mathcal{A}_J(S')$ is asserted. $\square$

Define the cellular direct image on the profile over I by actual $Ru_{S,*}$ on every complete inverse-image support S' , followed by its stratum restrictions and attaching data. Module actions are restricted along the algebra unit in Lemma 10.9. This gives a concrete meaning to μ_*^{cell} ; it is not a Kan extension on an undecorated interval.

Theorem 10.11 (Pairwise supported Morita comparison for integer cuts). *For $I \subset J$ in Proposition 10.3 and every allowed coarse support S with its complete inverse image S' , actual direct image induces equivalences*

$$\mathcal{C}_J(S') \xrightarrow[\text{Ru}_{S,*}]{\sim} \mathcal{C}_I(S), \quad \mathcal{C}_J^!(S') \xrightarrow[\text{Ru}_{S,*}]{\sim} \mathcal{C}_I^!(S).$$

They identify relative tensor and the supported action by the projection maps of Lemma 10.9, and Verdier partners by its pairing. Each coarse support-inclusion square and localization triangle is compared by its actual proper base-change map and mate. The retained ambient mapping spectra are compared by Lemma 10.10. The specified two-step refinement and attachment comparisons agree with actual composition. These functors identify the selected actual nearby and support realizations. This is a pairwise theorem with the displayed naturality maps; it does not assert an equivalence in an as-yet-unconstructed category of fully coherent six-operation/kernel profiles.

Proof. Use the module pullback

$$u_H^* M = H_{S'} \otimes_{u_S^* H_S}^L u_S^* M.$$

It is an actual extension-of-scalars functor, not ordinary sheaf pullback alone. It sends H_S to $H_{S'}$. On finite-cell modules the natural projection map identifies its adjunction unit as

$$M \longrightarrow Ru_{S,*} u_H^* M \simeq M \otimes_{H_S} Ru_{S,*} H_{S'} \simeq M.$$

The projection map is an equivalence on the unit and hence on finite cones and retracts; no projection assertion on all unbounded modules is needed. The algebra unit of Lemma 10.9 makes this a fully faithful functor. Its image is thick and contains $H_{S'}$. Theorem 10.6, which proved intrinsic geometric generation on both supports, therefore gives essential surjectivity. Its inverse is the stated actual direct image. The same argument, or the corresponding generator endomorphism comparison, proves the Morita equivalence rather than just equality of integrated complexes.

Proper Verdier duality commutes with $Ru_{S,*}$, with the actions induced by the algebra unit. It therefore gives the equivalence on $\mathcal{C}^!$, and carries $W_{S'}$ to W_S with its evaluation. Equivalently one may use the mixed projection comparison on $W_{S'} \otimes_{H_{S'}} u_H^* M$ and check it on $M = H_S$. Extension of scalars is strong monoidal; its inverse on these proved subcategories is consequently strong monoidal as well. Thus tensor, the mixed supported action and evaluation are intertwined.

For an inclusion of coarse supports, proper ordinary base change identifies restriction before and after direct image; its Verdier mate identifies costalk on the supported partners. For a coarse open/closed pair and its complete inverse image, composition of shriek images and proper base change give, in the ambient categories,

$$Ru_{*j'_!} j'^* M' \simeq j_! j^* Ru_* M', \quad Ru_* i'_* i'^* M' \simeq i_* i^* Ru_* M'.$$

The dual identities hold for the second localization triangle. They identify the units and counits themselves, since they are the mates of the same geometric squares. Applying the exact direct image to their fibre/cofibre sequences therefore identifies the boundary maps too. Lemma 10.10 compares the mapping spectra of these extension terms as well as their restrictions, so this compares the stated ambient gluing maps as well as the object values. A simultaneous higher profile statement is separated below. It does not assert an equivalence of the entire ambient constructible categories, nor preservation of the star subcategory by every extension term.

For $I \subset J \subset K$ the scheme maps compose as in Proposition 10.3. Derived module pullback has its canonical associativity cell, and the adjunction unit for a composite is the composite unit. The attachment identity and its two-step composition were proved in Equation (1) and Lemma 10.10. The above direct images and proper base-change transformations were all formed from this one geometric diagram, with the common generic fibre fixed. These constructions prove the stated two-step composition comparison. The finite cut poset supplies actual common refinements by union, and hence spans of the pairwise equivalences. We do not infer simultaneous higher support/tensor/Verdier/kernel coherence from the two-step calculation. In particular no unrestricted choice-independent profile equivalence is claimed here.

Finally the natural map $u_H^* N_I(F) \rightarrow N_J(F)$ is an equivalence on $F = Q$ and hence on the generated geometric source. Its adjoint is exactly proper nearby comparison; the supported version is its proved Verdier mate. This verifies the realization assertion for the same functors used in the cellular equivalence. $\square$

Target 10.12 (The simultaneous enhanced profile comparison). Construct the indexing correspondence category of cut models and complete inverse-image supports, together with its tensor and kernel operations, and a functor carrying it to the enhanced coefficient categories. Show that its structure maps are exactly the nearby, projection, evaluation, base-change and attachment maps constructed above. Verify the mixed pasting cells for support inclusions and refinements, their Verdier mates, all finite-arity tensor substitutions, and the diagonal pushforward cells in the same construction.

The full target remains broader than the pairwise theorem. Theorem 11.3 below proves the version with a fixed coarse support index, its fibrewise-cartesian ambient layers, and the specified minimal kernel diagrams. It does not construct an unrestricted correspondence category of all fine supports and kernels. The target is not a hypothesis on the coefficient category. The pairwise maps and their two-step identities provide its candidate structure maps. A list of commuting triangles is not a construction of this entire enhanced diagram.

10.6 The minimal kernel comparison, with support types retained

For an allowed support T let $\Delta_T : T \rightarrow T \times T$ be its diagonal. Kernels below carry the two actions of the pulled-back algebra H_T . The unit kernel is $\Delta_{T,*}H_T$ and its action is

$$M \longmapsto R p_{2,!}(K \otimes_{p_1^* H_T}^L p_1^* M).$$

For an open/closed pair $T = U \sqcup Z$ use also $\Delta_{T,*}j_!H_U$ and $\Delta_{T,*}i_*H_Z$.

Corollary 10.13 (Diagonal and normal-pair kernels descend). *The three kernels act respectively as the identity, $j_!j^*$ and i_*i^* , with the actual localization boundary. For complete inverse-image supports their proper pushforwards along $u_T \times u_T$ are the corresponding coarse kernels, with their module actions and boundary maps. Their actions and their Verdier mates therefore intertwine the cellular equivalence in Theorem 10.11.*

Proof. The diagonal projection formula over the actual algebra H_T reduces the three kernel actions to the indicated restriction/extension functors. Their triangle is the diagonal image of the ordinary localization triangle of H_T . Moreover $(u_T \times u_T)\Delta_{T'} = \Delta_T u_T$ and both u_T and the diagonals are proper. Hence the pushed unit is $\Delta_{T,*}R u_{T,*}H_{T'} = \Delta_{T,*}H_T$ by Lemma 10.9. The two other kernels follow from the open/closed identities in the preceding proof. Those identities preserve the boundary because its arbitrary extension class is carried by Equation (1); the two-refinement computation in Lemma 10.10 identifies its composites. Applying Verdier duality to this same localization triangle gives the asserted mates. Under the proved Morita presentations these are the associated bimodules and their actual tensor actions. $\square$

This is the promised minimal kernel interface. It does not provide a category of all relative-product kernels or prove general convolution closure and associativity. The projector terms have their recorded support types; they are not all endofunctors of a single cartesian coefficient category.

For example, when $m = 4$ the incomparable models with cuts $\{1\}$ and $\{2, 3\}$ have the actual common refinement $\{1, 2, 3\}$. The theorem gives pairwise comparisons along these proper maps. Supports on the incomparable models can be compared this way only when their complete inverse images agree on the common refinement; arbitrary different supports are not identified. It is compatible with a supplied finite group action on the whole geometric diagram by the equivariant-unit argument of Corollary 9.6. Constructing unsupplied lifts, adding arbitrary ramified base changes, retaining the infinite base Galois action, and identifying analytic/relative-root and integral realizations remain outside this fixed-thickness rational theorem.

11 Coherent descent of a fixed coarse support diagram

Fix a cut set B and let $\mathcal{P}_B$ be the finite poset of cut sets $J \supset B$, with arrows towards finer cuts. Let $\mathcal{L}_B$ be the poset of locally closed unions of strata of E_B . Our actual support index has objects (J, S) , $J \in \mathcal{P}_B$, $S \in \mathcal{L}_B$; its scheme at this object is $S_J = u_{JB}^{-1}S$. An inclusion $S \subset T$ gives $S_J \subset T_J$, and a refinement gives the proper map $S_K \rightarrow S_J$. These squares are cartesian by complete inverse image. Intersections, complements in open/closed decompositions, and all repeated inverse images are taken as schemes. This index does not add arbitrary new fine supports to the coarse support set.

At (J, S) use four types of coefficients:

$$\mathcal{A}_{J|B}(S_J), \quad \mathcal{A}_{J|B}^!(S_J) := \tilde{D}_{S_J} \mathcal{A}_{J|B}(S_J), \quad \mathcal{C}_J(S_J), \quad \mathcal{C}_J^!(S_J).$$

Here $\mathcal{A}_{B|B} = \mathcal{A}_B$. The ambient category is the intrinsic fibrewise-cartesian category of Lemma 10.10; arbitrary attachments between coarse strata are retained. A duality is typed as a functor from the opposite category. This avoids treating Verdier duality as covariant.

The support diagram consists of the following specified operations: ordinary restriction on the star ambient categories; $j_!$ and i_* for coarse open/closed decompositions; the two localization triangle functors into the ambient categories; their normalized Verdier conjugates on the supported ambient categories; relative tensor on $\mathcal{C}$, its action on $\mathcal{C}^!$, and the typed evaluations $M^{\vee H} \otimes_H M \rightarrow H$ and $M^{\vee H} \otimes_H (W \otimes_H M) \rightarrow W$. Include $R\Gamma$ and $R\Gamma_c$ with values in the fixed category $D(Q)$, and the specified global evaluation/trace pairings. Ambient Verdier duality includes its biduality and localization cells; no tensor-dualizability of every ambient object is assumed. Include the embeddings of these smaller categories into the ambient ones. Include the three diagonal/localization kernel actions of Corollary 10.13, with their displayed boundaries. Include the attaching mapping bifunctor $(M, V) \mapsto \text{Map}(M, L(V))$ with values in spectra. It is compared on the fine model by $\text{Map}(F_Z M, L'(V))$, not by requiring $L'(V)$ to be cartesian inside the contracted fibre. Star Rj_* is not asserted to preserve the fibrewise-cartesian ambient category. Its supported Verdier counterpart occurs in the dual localization triangle. These are specified kernel diagrams and actions, not a category of all kernels with arbitrary convolution. At B these operations and their cells are the actual enhanced module-sheaf operations used above.

Definition 11.1 (The operation diagram being transported). For the fixed B , take as types the four categories at each $S \in \mathcal{L}_B$ displayed above and their opposite categories, together with the fixed targets $D(Q)$ and spectra. Include the functor categories of arrows and cofiber sequences in each ambient type, with their evaluation functors; these type the localization triangle and its connecting arrow. For any list of input types and output type the operation object is the full infinity-category

$$\text{Fun}(\mathcal{D}_{c_1} \times \cdots \times \mathcal{D}_{c_n}, \mathcal{D}_d).$$

Substitution is functor composition. These operation objects define a Cat_∞ -enriched operation structure. They are not replaced by their maximal subgroupoids: noninvertible natural transformations, in particular localization arrows, are retained.

Fix its base subdiagram as follows. Its operations are precisely the support, localization, embedding, duality, relative tensor, global section, evaluation and attaching bifunctors enumerated above, and the three specified kernel actions. Include their identities and finite substitutions. Its cells are the actual adjunction units and counits, support composition/base-change cells, cofiber-sequence cells, biduality, tensor substitution and evaluation cells from the fixed base enhancement, with their simplices and substitutions. Equivalently one may take the subdiagram generated by this specified collection inside the

displayed functor infinity-categories. This fixes an operation diagram, not all operations on all coefficient categories. The three kernel objects themselves retain the separately specified proper-pushforward diagram, not an extra invertible kernel-category type.

At $J \supset B$ replace the four coefficient types by their stated fibrewise-cartesian versions on S_J , take opposites and triangle categories functorially, and keep $D(Q)$ and spectra fixed. Refinement acts by the module equivalences, their Verdier conjugates and the specified comparison transformations in the following theorem. Higher cells on the fine operation diagram will be transported from this fixed base subdiagram; they are not arbitrary preassigned fine cells.

Lemma 11.2 (Transport of a typed operation diagram). *Let a fixed enhanced diagram have coefficient categories $\mathcal{D}_c$ indexed by types c , with specified functors of finite arity and specified diagrams of natural transformations. Suppose $e_{j,c} : \mathcal{D}_c \simeq \mathcal{D}_{j,c}$ are equivalences forming a diagram in j , and each operation at j is equipped with a specified equivalence to the conjugate of the operation at the base. Then the conjugate diagram supplies coherent cells on these operations, including simultaneous mixed cells and their higher relations. This statement also applies to opposite categories and to operations valued in categories of triangles. It does not assert that arbitrary previously chosen cells coincide with the transported ones.*

Proof. For each list of input types and one output type, composition with e and its inverse gives an equivalence of functor categories

$$\mathrm{Fun}(\mathcal{D}_{c_1} \times \cdots \times \mathcal{D}_{c_n}, \mathcal{D}_d) \simeq \mathrm{Fun}(\mathcal{D}_{j,c_1} \times \cdots \times \mathcal{D}_{j,c_n}, \mathcal{D}_{j,d}).$$

Use the unit and counit of these equivalences in substitutions. Their triangle identities identify successive substitutions, and the associativity of functor composition supplies all iterated substitutions. Equivalently these equivalences identify the colored operation objects, which are the displayed full functor infinity-categories, with composition given by substitution. Noninvertible natural transformations are arrows in these categories and remain present throughout. Transport the whole base diagram through this identification, then through the specified equivalences of operations. This transports simplices, not just objects or edges. The fibre of an equivalence of mapping spaces over a specified simplex is contractible, so this introduces no additional independent choices of higher cells. The same construction in functor categories handles triangles, and opposites handle the contravariant operations. Functorial equivalences $e_{j,c}$ make the construction simultaneous in j . No claim about a contractible space of all possible geometric enhancements is needed. $\square$

Theorem 11.3 (Coherent refinement descent with a fixed coarse support index). *For the fixed-thickness integer-cut family, the preceding typed support diagram admits a coherent enhancement under refinement in $\mathcal{P}_B$. Its star transition is actual derived module pullback and its inverse is actual proper direct image. On supported types the transition is its normalized Verdier conjugate, again inverse to proper direct image. It retains the ambient localization terms, the specified support maps, tensor and evaluation, and the three minimal kernel comparisons. The higher mixed cells are those transported from the fixed base operation diagram of Definition 11.1. The listed elementary support/refinement squares are identified with the actual base-change mates; no equality with every independently prescribed fine enhancement is claimed. All these refinement edges can be inverted coherently. This proves descent for this fixed coarse support index, not equivalence with all fine ambient categories or with a profile of arbitrary fine supports.*

Proof. Write $F_{JB} = H_J \otimes_{u_{JB}^* H_B} u_{JB}^*(-)$, restricted to S_J where appropriate. The nearby algebra maps in Lemma 10.9 are obtained from the generic inverse-image diagram and its

adjunction counits. Thus for a string of actual model maps their composites are the mates of that same composite counit. Constructing mates on the functor categories of strings gives a diagram of algebra pullbacks, rather than independently choosing pairwise algebra maps. Derived tensor associativity then defines $F_{KB} \simeq F_{KJ}F_{JB}$ on all strings. The threefold associativity cell is the associativity cell for this single iterated tensor product; the pentagon and its higher versions are those of that tensor product. These are the requisite functorial equivalences $e_{J,c}$.

Lemma 10.10 proves that F_{JB} is an equivalence $\mathcal{A}_B(S) \simeq \mathcal{A}_{J|B}(S_J)$. Its restriction gives the equivalence on $\mathcal{C}$. On the supported types define the covariant functor by the typed composition

$$\mathcal{A}_B^!(S) \xrightarrow{\tilde{D}_S} \mathcal{A}_B(S)^{\text{op}} \xrightarrow{F_{JB}^{\text{op}}} \mathcal{A}_{J|B}(S_J)^{\text{op}} \xrightarrow{\tilde{D}_{S_J}} \mathcal{A}_{J|B}^!(S_J).$$

We denote it by $F_{JB}^!$; its restriction gives the same formula on $\mathcal{C}^!$. Thus the middle functor is explicitly the opposite of the star functor. Proper duality identifies its inverse with $Ru_{JB,*}$, by the explicit evaluation/trace pairing of Lemma 10.9. This also defines its composition cells; ordinary pullback is not substituted for this supported transition.

We specify the conjugation maps for the operation diagram. For a support inclusion $k : S \rightarrow T$ they are

$$k^*F_{JB} \simeq F_{JB}k^*, \quad k^!F_{JB}^! \simeq F_{JB}^!k^!.$$

The first is inverse-image/tensor associativity on the actual cartesian square, and the second is its Verdier mate. For a coarse open/closed pair, $F_{JB}j_! \simeq j_{J,!}F_{JB}$ and $F_{JB}i_* \simeq i_{J,*}F_{JB}$ follow from extension by zero and closed base change. The map

$$F_{JB}Rj_* \xrightarrow{\chi_{JB}} Rj_{J,*}F_U$$

is the mate of the open counit, but need not be invertible. Its fine target can be supported at the ends of a contracted fibre and need not belong to $\mathcal{A}_{J|B}$. We therefore compare the attaching mapping bifunctor instead: the actual proper comparison $G_Z L'(V) \simeq L(V)$ and the module adjunction give

$$\text{Map}(F_Z M, L'(V)) \simeq \text{Map}(M, G_Z L'(V)) \simeq \text{Map}(M, L(V)).$$

These are equivalences of bifunctors with values in spectra. Lemma 10.10 proves that their mate sends an attachment to its actual pullback via the closed restriction of χ_{JB} . Thus the required operation is conjugated even though the external link itself is not. The dual ambient localization operations are obtained by Verdier conjugation of $j_!$ and i_* , so no closure under star Rj_* has been inserted. The localization triangle is transported as the cofiber of its specified unit/counit. Tensor conjugation is the associativity map of relative tensor; the mixed supported action is the projection mate specified in Lemma 10.9. Evaluation conjugation is its displayed trace pairing. The global-section operations are conjugated by $R\Gamma(S_J, -) = R\Gamma(S, Ru_*(-))$ and $R\Gamma_c(S_J, -) = R\Gamma_c(S, Ru_*(-))$, since u is proper, followed by the proved unit equivalence on the corresponding type. Their scalar target has identity transition. This also types the global pairings without assuming an equivalence of unrestricted ordinary sheaf categories. Each minimal kernel is the diagonal image of one of these localization terms. The identity $(u \times u)\Delta_J = \Delta_B u$ and composition of direct images give its specified pushforward conjugation, including the action and boundary. More precisely these kernels are the finite diagram $\Delta_* j_! H_U \rightarrow \Delta_* H_T \rightarrow \Delta_* i_* H_Z$. At a finer model they are its actual module pullbacks before applying that model's diagonal. Proper pushforward of the diagonal diagram uses $Ru_* F_u \simeq \text{id}$ on the three ambient terms. The unit for a composite is the composite unit, so the resulting kernel comparisons on every refinement string form

a cartesian section of the proper-pushforward diagram of kernel objects. Their actions are the already transported localization operations. We do not apply Lemma 11.2 to the entire kernel categories, nor claim full faithfulness of forgetting a diagonal bimodule action. No kernel pullback equivalence on an unrestricted product category is used.

These explicit conjugations verify the hypotheses of Lemma 11.2. Apply it to the entire base operation diagram at once. This constructs the higher mixed cells, not just checks of its separate triangles. For example, pasting a support square with a refinement square gives on star restriction the same inverse-image/tensor associativity map in either order. Taking the counit mate gives exactly the mixed attachment square involving χ (without declaring it invertible); dualizing gives the costalk square; taking the cofiber gives its boundary square. The projection and evaluation formulas identify their tensor and Verdier variants with these transported cells. Thus the elementary geometric squares have their original arrows. Higher mixed cells are the transported cells just constructed; we do not claim uniqueness of an unrelated enhancement of all six operations.

Finally the coefficient/operation diagram, whose refinement arrows are equivalences, factors through the localization of its index at these arrows. This is the universal property of categorical localization, applied to the entire typed diagram. Its inverse arrows are the actual Ru_* with the proved units and counits. The poset $\mathcal{P}_B$ has initial object B and terminal full cut; unions give actual common refinements. The resulting coefficient and operation identifications are therefore independent of the path within this anchored index, with the constructed higher cells. The selected kernel objects have the compatible cartesian section just described; no ambient kernel-category arrow is inverted. Changing the anchor to a finer model compares the old diagram on the corresponding fibrewise-cartesian subcategories, not with the whole new ambient layer. $\square$

Remark 11.4 (A loop is a new coefficient problem). The forest argument in Theorem 10.6 is essential. For a constant diagram of coefficient categories on a graph with one cycle, take the unit at every vertex and identity attachments except one scalar automorphism $\lambda \in Q^\times$, $\lambda \neq 1$. Every branch map is invertible, but the product around the cycle is λ . This object is not generated thickly by the global unit: finite extensions, cones and retracts of the unit have only generalized eigenvalue 1 for the cycle monodromy, whereas this object has monodromy λ . This is an obstruction for the categorical graph model, not a claimed geometric realization on every nodal curve. For general skeleta one must retain the diagram and its global monodromy, or prove a geometric global-unipotence restriction and its operation closure. The anchored theorem above makes neither claim.

12 Relevant prior results and the remaining comparison

The logarithmic comparison in Theorem 3.2 is an application of Fujiwara–Kato, not a new general invariance theorem. It gives a useful upstream coefficient source with actual refinement units. To obtain the target datum one must still descend the base inertia, compare the ordinary/nearby and analytic realization functors, and identify their supported partners and attaching maps.

For background we use the PDF of Achinger–Ogus, [arXiv:1802.02234v2](#), §6.3, Theorems 6.3.3–6.3.4 [6]. These discuss exterior nearby cohomology and connecting/monodromy information; the complex must not be replaced by its cohomology with zero attaching maps. All numbering here follows that PDF, not the experimental HTML rendering. No Achinger–Ogus theorem is used as a finite-coefficient, adic or module input in this paper. The local proof instead uses Illusie’s Theorem 8.3, Corollary 8.4 and §5, with the coefficient passage proved above.

There is a separate coherent-sheaf result: Scherotzke–Sibilla–Talpo [7, Theorem B/Proposition 3.9] prove derived invariance for rational-weight coherent parabolic sheaves under the stated smooth normal-crossings log blow-ups. These are not constructible ℓ -adic sheaves. In addition our original thickness-two model is singular and its monoid P is not free. Thus their theorem is relevant precedent for a Morita-valued invariant, but cannot be substituted for the coefficient or supported comparison of this paper.

Curry’s cellular theory provides a related comparison: [8, §6.3, Theorem 6.12; Theorem 7.9 and §7.4] discusses subdivision invariance of cellular cohomology and derived sheaf-cosheaf equivalence with Verdier duality. These are background mechanisms on cell complexes; they do not supply the étale nearby algebra, nontrivial branch cochain modules, complete exceptional support or actual proper comparison used here. We use no poset Kan-extension proof from that paper as an input, so the present arguments do not depend on its original Kan-extension argument or its subsequent erratum.

For the categorical assembly, Haine–Porta–Teyssier [9] prove exodromy and coarsening results under their exodromic stratified-topos hypotheses. Such results can organize refinement after the actual coefficient categories, admissibility and appropriate stratified topos have been identified. We have not checked those hypotheses for the logarithmic/adic realization used here, so no exodromy theorem is invoked to certify the present cellular descent.

The concrete endpoint map is now computed in Theorem 9.2 using the actual Kummer classes, and the distinct support restriction $\kappa^!$ is identified in Theorem 9.4. These geometric results were not inferred from the finite logarithmic adjunction unit. Relating their full base-inertia descent and finite-root trace realizations to that logarithmic comparison still requires a separate comparison square. The support categories, pairwise minimal kernel transport and stated two-step cells for integer cuts over a fixed base are supplied by Theorem 10.11 and Corollary 10.13. No exodromy or coherent-sheaf invariance result is used to bypass these actual module-sheaf arguments.

13 The common coefficient interface and one-modification bridge

We use the corollary “The geometric unipotent interface for refinement” and the following shared-coordinate remark of the fixed 16 September 2026 companion version [1]. A complete source snapshot is supplied in `companion_sources`; its manifest records the SHA-256 and the stable source labels `cor:frag-refinement-unipotent-interface` and `rem:frag-refinement-coordinate-`. They are proved there from the fixed-node theorem and unit generation, without using the refinement theorem of this paper. Thus the dependency is one-way. Set $e = 0$, identify the two uniformizers $t = \pi$, and fix one compatible family $t_n^{\ell^n} = t$, $t_{n+1}^\ell = t_n$ in $\bar{K}$. Retain the geometric rational unipotent source $\mathcal{U}_\Gamma$ of that corollary, where $\Gamma = \mathbf{Z}_\ell(1)$. We use its right-module convention; commutativity identifies this with the module convention used above. Its actual associated-sheaf functor is

$$V \mapsto F_V \in \mathcal{U}^b(\mathbf{G}_{m,\bar{K}}), \quad A_\Gamma = K_\Gamma(Q).$$

It uses the torsors $z_n^{\ell^n} = x$, not a coefficient selected from the isomorphism class of its cohomology. The cited interface gives $\mathcal{U}_\Gamma \simeq \mathcal{U}^b(\mathbf{G}_{m,\bar{K}}) \simeq \text{Perf}(A_\Gamma)$, with coefficient tensor corresponding to $\otimes_{A_\Gamma}$. No such claim is made for the entire source of the fixed-node theorem.

Proposition 13.1 (The actual one-modification realization bridge). *For $X : xy = t^2$ and the modification of Proposition 2.1, put $W(V) = \kappa^! R\Psi_{X'} F_V$ and $N(V) = \kappa^* R\Psi_{X'} F_V$. The functor $V \mapsto F_V \mapsto N(V)$ identifies $\mathcal{U}_\Gamma$ with the intrinsic star category $\mathcal{C}_*$. There are specified algebra and module equivalences*

$$\theta : A_\Gamma \xrightarrow{\sim} R\Gamma(E, H), \quad \alpha_V : K_\Gamma(V) \xrightarrow{\sim} R\Gamma(E, N(V)), \quad H = N(Q),$$

and the supported comparison is

$$\beta_V : R\Gamma(E, W(V)) \xrightarrow{\sim} D_Q K_\Gamma(V^\vee)(-1)[-2].$$

They preserve the actual coefficient actions, relative tensor, evaluation pairing and endpoint localization attachments. At either new node q , with $s_q : q \rightarrow E$ and $i_q = \kappa s_q$, the local supported value of the first paper is

$$i_q^! R\Psi_{X'} F_V = s_q^! W(V),$$

with the corresponding coordinate transport of V . It is not the ordinary stalk $s_q^* W(V)$.

Proof. Source and star map. Use exactly the associated torsors in the cited interface. Define α_V to be the composite

$$K_\Gamma(V) \longrightarrow R\Gamma(\mathbf{G}_{m, \bar{K}}, F_V) \longrightarrow B(F_V) \longrightarrow R\Gamma(E, N(V)).$$

The first arrow is the associated-sheaf cochain map from that interface; the second is actual nearby restriction; the third is proper nearby comparison on the complete inverse-image support. The last two are those of Theorem 9.5, not an application of a thickness-one theorem at the old node. At the unit their maps send $[x]$ to the primitive relative Kummer class, including at $xy = t^2$; there is no factor of two. They are algebra maps by Lemma 10.9, and the first map has the cochain algebra structure specified in the interface. This gives θ . Their coefficient versions are module maps, and their equivalence locus is thick. Unit generation therefore gives α_V on the entire common subspace. Theorem 9.2 identifies its actual sheaf image as $\mathcal{C}_*$.

Coordinate and attachment maps. On $U = E^\circ$ use $u = x/t$ and at the other end $v = u^{-1}$. At every finite Kummer level the maps are literally

$$z_n \longmapsto z_n/t_n, \quad z_n/t_n \longmapsto t_n/z_n.$$

The first commutes with the deck action, while the second intertwines it with $\zeta \mapsto \zeta^{-1}$. Thus the local coefficient in the v coordinate is the pullback of V along this inverse group map; it is not identified with V by forgetting its action. These are exactly the torsor maps fixed in [1]. The degree-one signs are consequences, not definitions, of these maps.

Write $L_q = i_q^{E,*} Rj_*$ for the link inside E (here $i_q^E = s_q$). The actual open unit and restriction of sections give a square

$$\begin{array}{ccc} R\Gamma(E, N(V)) & \longrightarrow & R\Gamma(U, j^* N(V)) \\ \downarrow & & \downarrow \\ s_q^* N(V) & \longrightarrow & L_q(j^* N(V)). \end{array}$$

All four arrows are restrictions or the open-adjunction unit. Under α_V and the displayed torsor identifications, its bottom arrow is the link map of Theorem 9.2. Its composite with the torus-to-node star map is induced by exactly the torsors used in the first paper's local realization. Thus that paper supplies the local coefficient identification, while the present paper supplies the link and localization arrow. The unit computation identifies these maps on cochains; their module naturality and exactness extend it to F_V , without splitting its extensions. Recollement reconstructs the same attaching triple; its localization boundary is the connecting arrow of this very unit, so no new boundary is chosen from a cohomology formula.

Supported map and evaluation. Define β_V by the actual nearby-duality and proper-duality pairing, followed by the dual of α_V :

$$R\Gamma(E, W(V)) \longrightarrow D_Q R\Gamma(E, N(V^\vee))(-1)[-2] \xrightarrow{D_Q \alpha_V^\vee(-1)[-2]} D_Q K_\Gamma(V^\vee)(-1)[-2].$$

The first arrow is an equivalence by Theorems 9.4–9.5 and properness of E . Its action and trace are the specified H -linear pairing of Lemma 10.9. In particular

$$\begin{array}{ccc} R\Gamma(E, N(V^\vee)) \otimes_Q R\Gamma(E, W(V)) & \longrightarrow & Q(-1)[-2] \\ \downarrow \alpha_{V^\vee}^{-1} \otimes \beta_V & & \parallel \\ K_\Gamma(V^\vee) \otimes_Q D_Q K_\Gamma(V^\vee)(-1)[-2] & \longrightarrow & Q(-1)[-2] \end{array}$$

commutes by construction. Moving the algebra action between the inputs gives the same pairing, so this also identifies the supported A_Γ -module action. Relative tensor is the balanced cochain map on the first side and the actual H -module tensor on the second; it is an equivalence on two units and then on both thick closures.

Each new node has chart $zw = t$. With the above torsor transport, the first paper applies there with $e = 0$ and its fixed rank-one carrier. Its tame geometric comparison agrees with the ordinary nearby complex used here on this selected Kummer source, by the wild-triviality comparison in that paper. Its supported arrow is specified by the same evaluation and geometric trace; taking the Verdier mate of the just identified local star map gives the node support comparison from the same sheaf-level pairing that defines β . It is not obtained by restricting an integrated complex or by choosing a new dual isomorphism. Composition of exceptional restrictions gives $i_q^! = s_q^! \kappa^!$. An auxiliary compactification has boundary away from q , so it does not alter this costalk. The old thickness-two node and the positive-dimensional U enter through this paper's Kummer calculation and proper comparison only; the first paper's node theorem is not extended.

Coefficient levels. The torsor identities and evaluation maps above are first defined at finite level and commute with root transition and scalar change. For a source coefficient at precision r , use its finite-action presentation in the first paper; no common finite root bound for all r is required. Its adic maps are those specified there and in Lemma 4.1. Rationalizing gives the actual natural transformations used above. Unit generation is applied at this rational stage only. This does not promote the rational Morita interface to the whole integral coefficient source. $\square$

Example 13.2 (Zero endpoint stalk and nonzero node support). For $V = Q$, Theorem 9.4 gives $W(Q) = j_! Q_U$. Hence $s_q^* W(Q) = 0$, but localization gives

$$s_q^! W(Q) \simeq L_q(Q_U)[-1] \simeq D_Q K_\Gamma(Q)(-1)[-2].$$

It has Q in degree one and $Q(-1)$ in degree two. This is the nonzero node supported value of the first paper. For general V the same formula uses $L_q(j^* N(V))[-1]$ with its actual action and coordinate transport. In particular $s_q^! N(V) = 0$ at an outside node is a statement about the star object, not about $W(V)$. At an internal node of a longer chain one must instead use the branchwise calculation in Example 10.7.

Corollary 13.3 (The local-realization bridge in the anchored diagram). *Fix m , the coordinate x and the compatible base roots of the preceding interface. The actual functors*

$$V \longmapsto N_{I,S}(V) = k_S^* \kappa_I^* R\Psi_{X_I} F_V, \quad V \longmapsto W_{I,S}(V) = (\kappa_I k_S)^! R\Psi_{X_I} F_V$$

form the coefficient realization and its supported partner in the anchored diagram of Theorem 11.3. Here S denotes an allowed support on E_I , and comparisons under refinement always use its complete inverse image. Put $A_{I,S} = R\Gamma(S, H_S)$, with its actual map from A_Γ . The specified cochain and pairing maps give

$$K_\Gamma(V) \otimes_{A_\Gamma}^L A_{I,S} \xrightarrow{\sim} R\Gamma(S, N_{I,S}(V)),$$

$$R\Gamma(S, W_{I,S}(V)) \xrightarrow{\sim} D_Q R\Gamma_c(S, N_{I,S}(V^\vee))(-1)[-2].$$

For $S = E_I$ the first algebra map is an equivalence and these become

$$R\Gamma(E_I, N_I(V)) \simeq K_\Gamma(V), \quad R\Gamma(E_I, W_I(V)) \simeq D_Q K_\Gamma(V^\vee)(-1)[-2].$$

The comparisons retain module actions, evaluation and the actual mixed support/refinement attachment squares. The same transported higher cells as in the anchored theorem organize these maps; no separate choice of a comparison on each model is made.

Proof. Use the single associated-sheaf functor on the common generic torus. The algebra map $A_\Gamma \rightarrow A_{I,S}$ is its unit cochain map followed by nearby restriction to S . The balanced coefficient map in the first display is evaluation of this same natural transformation. It is an equivalence on $V = Q$, and both functors are exact; hence it is an equivalence on the proved thick unit subsource. This does not identify the source with $\mathcal{C}_I(S)$ for every disconnected support; the displayed scalar extension is essential.

For $I \subset J$ the square of these transformations is induced by

$$F_{JI} N_{I,S}(V) \longrightarrow N_{J,S'}(V).$$

Its map is the nearby mate of the generic identity; it is the unit identification at Q and an equivalence throughout the same thick source. For an inclusion $S \subset T$, restricting this mate gives the mate on S because the generic pullback and open counit defining it are the same after the cartesian support square. This proves the mixed square for the original arrows, including the attachment bifunctor through $G_Z L' = L$. The coordinate equations on every gap are $x = t^a u$ and its inverse branch, so the torsor maps of Proposition 13.1 apply unchanged, with inverse deck action retained on that branch.

Define the second display by the actual nearby evaluation followed by the Verdier global-section pairing. For a nonproper S , Verdier duality says $R\Gamma(S, D_S M) = D_Q R\Gamma_c(S, M)$, which explains the compact support on the right. Its H_S -action and proper refinement map are the pairing and projection mates already used in the anchored theorem. Thus mixed costalk/refinement squares are the Verdier mates of the just proved star squares. One cannot replace $R\Gamma_c(S, -)$ by $R\Gamma(S, -)$ in this formula for arbitrary S .

For the full E_I , proper comparison with the original node and Proposition 10.1 identify $A_\Gamma \rightarrow R\Gamma(E_I, H_I)$ by the primitive class $[x]$. Exactness and generation give its coefficient version. The support E_I is proper, so $\Gamma_c = \Gamma$ there, proving the final display. At new thickness-one nodes this agrees with the first paper's supported interface as a costalk, as in Example 13.2. Higher-thickness gaps use this paper's calculation, not an extension of the first paper's model domain.

Finally add the fixed source $\mathcal{U}_\Gamma$ as an extra type with identity transition to the operation diagram in Lemma 11.2. Add N , its supported partner, the specified cochain maps and evaluation. The conjugation maps for these added operations are precisely the nearby mates just identified; their support and attachment squares were checked above. Transporting this enlarged diagram constructs all its higher cells simultaneously with those of Theorem 11.3. The finite torsor maps, selected adic systems and rationalization are those of Lemma 4.1; the use of thick generation here is only on the rational common source. $\square$

Proposition 13.1 concerns one modification; Corollary 13.3 extends its specified arrows to the fixed-cut anchored diagrams on the same rational unipotent subsource. The resulting coherent statement is restricted to that anchored operation diagram; it does not close the unrestricted Target 10.12 or extend the first paper's finite-carrier theorem to arbitrary models.

The proved scope includes geometric unipotent generation with supplied finite descent, actual algebraic normal-pair reconstruction and Verdier duality, its fixed-pair kernel actions, and the single-modification nearby star/module and complete-support comparison. The geometric admissible image is now identified, and is smaller than the full exceptional constructible category. The fixed-thickness theorem supplies pairwise comparisons of the actual ambient gluing maps, Verdier partners and minimal kernels under integer cuts, with the specified support squares and two-step cells. Their simultaneous organization on a fixed coarse support index is Theorem 11.3, with the natural bridge of Corollary 13.3. The unrestricted Target 10.12 and the loop-monodromy extension are not consequences of this anchored theorem. General model/base-change independence, full relative-product convolution, base-inertia/logarithmic comparison, analytic compatibility beyond the identified local-node interface, and integral supported refinement remain open.

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